

27. Superconductivity

27.1 Introduction

Kamerlingh Onnes (1911) found that mercury loses all electrical resistance below a critical temperature of 4 K. In copper, a very good conductor at temperatures of 77 K, electrical currents decay with a characteristic time of $2 \cdot 10^{-13}$ s. But a superconductor is qualitatively better. Currents in a superconductor have been seen to persist for over a year. Since Kamerlingh Onnes' initial discovery, most metals and alloys have been found to exhibit superconductivity at some temperature, so the phenomenon is quite universal.

The phenomenon of superconductivity posed for many years a serious challenge to quantum mechanics. Perhaps the dense combination of many electrons really could not be described by the same equations that had been deduced from experiments on dilute systems; perhaps quantum mechanics was just an approximation to more complicated many-body equations, and superconductivity was the key to finding the true form. Sommerfeld and Bethe (1933) wrote that “[d]espite failure up to now, we may assert that superconductivity will be solved on the basis of our present quantum mechanical understanding.” This assertion remained a matter of faith for another 25 years as one microscopic theory after another collapsed until Bardeen, Cooper, and Schrieffer (1957) created a compelling quantum-mechanical model.

An important idea to grasp in understanding superconductivity is that it really is a macroscopic phenomenon, and corresponds to a definite macroscopic state of matter. The important feature of a superconductor is not how it responds to external electrical fields, but how it responds to external magnetic fields. Meissner and Ochsenfeld (1933) found that magnetic flux is completely expelled from a superconductor. Suppose one has a loop with a current flowing in it. Magnetic flux lines must thread the loop, as in Figure 27.1. In order for the current to diminish,

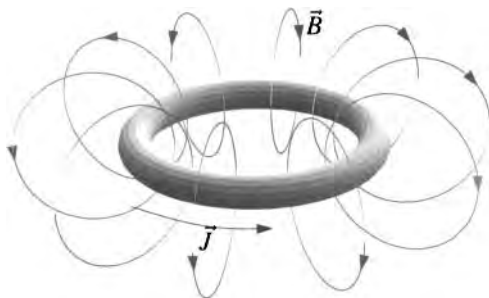


Figure 27.1. The flux lines threading a current loop have to pass through the current if the current is to diminish. But for a superconductor, that is impossible, and the current persists.

the total magnetic field it creates would have to be reduced as well. In the course of the reduction, a flux line would have to pass through the current loop, but for a superconductor that is nearly impossible. It is for this reason that currents in superconductors are so persistent. The impurity scattering sites in the metal and the collisions of electrons with phonons do not go away, but whenever an electron is slowed down by such a collision, a flux line heads toward the metal surface. The metal repels it, and in so doing pushes the tardy electron back up to speed.

27.2 Phenomenology of Superconductivity

Perfect Diamagnets. When a superconductor is placed in an external field $\vec{H}$, all the flux lines are expelled from it so $\vec{B}$ is zero in its interior. The magnetic permeability μ described by Eq. (24.6) is zero, and the superconductor is a perfect diamagnet. This perfect diamagnetism can be explained phenomenologically by supposing superconductors to be solids in which electrons accelerate in the presence of electric fields without displaying any damping. Like magnets and dielectric media, they are captured by proposing a relation between electrical field and current, which is

$$m\dot{\vec{v}} = -e\vec{E} \quad (27.1)$$

$$\Rightarrow \frac{\partial \vec{j}}{\partial t} = \frac{ne^2}{m}\vec{E} \quad (27.2)$$

$$\Rightarrow \frac{\partial}{\partial t} \vec{\nabla} \times \vec{\nabla} \times \frac{\vec{B}}{\mu} = \frac{4\pi ne^2}{mc} \vec{\nabla} \times \vec{E} \quad (27.3)$$

Using Eq. (24.5) and restricting attention to slowly varying currents so that $\partial \vec{B}/\partial t$ can be ignored.

$$\Rightarrow \vec{\nabla} \times \vec{\nabla} \times \frac{\vec{B}}{\mu} = -\frac{4\pi ne^2}{mc^2} (\vec{B} - \vec{B}_0). \quad (27.4)$$

Using again Eq. (20.5b) and integrating in time, obtaining a constant of integration $\vec{B}_0$.

This line of argument cannot determine the constant field $\vec{B}_0$. Because of Meissner and Ochsenfeld's observation that all magnetic fields are expelled from a superconductor, London (1961) proposed that superconductivity could be explained by setting $\vec{B}_0$ to zero to obtain

$$\vec{B} + \lambda_L^2 \vec{\nabla} \times \vec{\nabla} \times \vec{B} = 0 \quad (27.5)$$

where the *London penetration depth* is

$$\lambda_L = \sqrt{\frac{mc^2}{4\pi\mu ne^2}}. \quad (27.6)$$

The assumption is that any features of magnetic response not captured explicitly by $\vec{j}$ are captured by a coefficient μ .

Equation (27.5) is called the *London equation*. The length λ_L gets its name from the solutions of Eq. (27.5). Consider a superconductor which occupies all of space for $z > 0$. Look for solutions of Eq. (27.5) which are independent of x and y . Rewrite Eq. (27.5) as

$$\vec{B} + \lambda_L^2 \left(\vec{\nabla} (\vec{\nabla} \cdot \vec{B}) - \nabla^2 \vec{B} \right) = 0. \quad (27.7)$$

Taking the dot product of Eq. (27.7) with $\hat{z}$ gives

$$B_z = 0. \quad \text{Because } B \text{ is independent of } x \text{ and } y, \text{ all the } x \text{ and } y \text{ derivatives vanish.} \quad (27.8)$$

The first conclusion is then that the normal component of the magnetic induction always vanishes at the surface of a superconductor. Taking next a dot product with respect to $\hat{x}$ gives

$$B_x = \lambda_L^2 \frac{\partial^2 B_x}{\partial z^2} \Rightarrow B_x \propto e^{-z/\lambda_L}. \quad (27.9)$$

The second conclusion is that a component of the induction parallel to a superconducting surface dies off exponentially over length λ_L . This decay is the origin of the Meissner effect. Because the current within the superconductor is related to the induction by Eq. (24.3), one obtains two additional important conclusions:

1. The superconductor sustains currents in equilibrium.
2. These currents inhabit a strip of width λ_L near the superconductor's surface, and they vanish everywhere else.

27.2.1 Phenomenological Free Energy

Trying to explain superconductivity with Eq. (27.2) as the starting point creates two difficulties. First, the constant time-independent magnetic field $\vec{B}_0$ needs to be eliminated without explanation in the midst of the analysis. Second, it is not the most general relation leading to superconductivity. Both difficulties can be counteracted by recasting the phenomenological theory in terms of a free energy, rather than a linear relation between current and field.

Consider a material without macroscopic magnetic fields in equilibrium. Weinberg (1986) emphasizes the economy of beginning with a free energy whose magnetic-field-dependent part takes the form

$$\mathcal{F} = \int d\vec{r} d\vec{r}' \sum_{\alpha\beta} \frac{1}{2} A_\alpha(\vec{r}) G_{\alpha\beta}(\vec{r} - \vec{r}') A_\beta(\vec{r}') + \delta(\vec{r} - \vec{r}') \frac{1}{8\pi} \vec{B}(\vec{r}) \cdot \vec{B}(\vec{r}'). \quad (27.10)$$

G is an arbitrary phenomenological function. Because it is a function of $\vec{r} - \vec{r}'$, it describes a macroscopically homogeneous medium.

Equation (27.10) differs from the previous magnetic free energy (24.25) in two respects. It is less general, because Eq. (27.10) is only quadratic in $\vec{A}$, and therefore only suitable for small deviations from equilibrium. On the other hand, this form puts forward $\vec{A}$ as the primary variable rather than $\vec{B}$. A technical point that should be mentioned is that Eq. (27.10) is therefore not gauge-invariant. This deficiency can be remedied by replacing $\vec{A}$ wherever it appears by $\vec{A} - \vec{\nabla}\phi$, where a gauge change $\vec{A} \rightarrow \vec{A} + \vec{\nabla}\chi$ sends ϕ to $\phi + \chi$. The final results will all be expressed in terms of $\vec{B} = \vec{\nabla} \times \vec{A}$, and because $\vec{\nabla} \times \vec{\nabla}\phi = 0$ is a vector identity there is no need to carry ϕ along.

The minima of this free energy describe equilibrium states. Requiring the functional derivative of the free energy to be zero with respect to the vector potential gives

$$[\vec{\nabla} \times \vec{\nabla} \times \frac{\vec{A}(\vec{r})}{4\pi}]_{\alpha} = - \int d\vec{r}' \sum_{\beta} G_{\alpha\beta}(\vec{r} - \vec{r}') A_{\beta}(\vec{r}') \quad (27.11)$$

Write $\vec{B} = \vec{\nabla} \times \vec{A}$ before taking derivatives of Eq. (27.10).

$$\Rightarrow j_{\alpha}(\vec{r}) = \frac{c}{4\pi} [\vec{\nabla} \times \vec{B}]_{\alpha} = -c \int d\vec{r}' \sum_{\beta} G_{\alpha\beta}(\vec{r} - \vec{r}') A_{\beta}(\vec{r}'). \quad (27.12)$$

In Fourier space, Eq. (27.11) becomes

$$\sum_{\beta} \left\{ G_{\alpha\beta}(\vec{k}) + \frac{1}{4\pi} (k^2 \delta_{\alpha\beta} - k_{\alpha} k_{\beta}) \right\} A_{\beta} = 0. \quad (27.13)$$

A system displays paramagnetic or diamagnetic behavior with permeability μ if in the limit as $k \rightarrow 0$,

$$G_{\alpha\beta} \rightarrow \left(\frac{1}{\mu} - 1 \right) \frac{1}{4\pi} [k^2 \delta_{\alpha\beta} - k_{\alpha} k_{\beta}]. \quad \text{Not obvious—obtained by working backwards from the desired result.} \quad (27.14)$$

To see why, note that if G is taken to have this form, then the term involving G in the free energy is

$$\frac{1/\mu - 1}{8\pi} \int d\vec{r} d\vec{r}' \delta(\vec{r} - \vec{r}') \vec{A} \cdot \vec{\nabla} \times \vec{\nabla} \times \vec{A} \quad \text{Write } \sum_{\beta} [k^2 \delta_{\alpha\beta} - k_{\alpha} k_{\beta}] A_{\beta} = -[\vec{k} \times \vec{k} \times \vec{A}]_{\alpha}. \quad (27.15)$$

$$= \frac{1/\mu - 1}{8\pi} \int d\vec{r} B^2(\vec{r}). \quad \text{Use the identity in Eq. (24.21) to integrate by parts and turn the leftmost } \vec{A} \text{ into } \vec{B}. \quad (27.16)$$

Adding Eq. (27.16) to the second term in Eq. (27.10) gives a free energy

$$\mathcal{F} = \frac{1}{8\pi\mu} \int d\vec{r} B^2(\vec{r}), \quad (27.17)$$

which describes a conventional permeable magnetic substance.

A superconductor is produced by adding any term to $G_{\alpha\beta}$ that destroys the existence of solutions with spatially uniform magnetic fields. For example, suppose that

$$\lim_{k \rightarrow 0} G_{\alpha\beta} = \frac{1}{4\pi\lambda_L^2} \delta_{\alpha\beta}. \quad (27.18)$$

In this event, the free energy takes the form

$$\frac{1}{8\pi} \int d\vec{r} \frac{1}{\lambda_L^2} A^2(\vec{r}) + |\vec{\nabla} \times \vec{A}|^2, \quad (27.19)$$

which has an extremum when

$$\frac{1}{\lambda_L^2} \vec{A} + \vec{\nabla} \times \vec{\nabla} \times \vec{A} = 0. \quad (27.20)$$

Taking the curl of Eq. (27.20) to ensure that the final results are gauge-invariant gives

$$\vec{B} + \lambda_L^2 \vec{\nabla} \times \vec{\nabla} \times \vec{B} = 0, \quad (27.21)$$

recovering Eq. (27.5).

Equation (27.21) is not quite as general as Eq. (27.5) because the first involves $\vec{B}$ while the second involves $\vec{B}/\mu$. A magnetic permeability different from unity can be introduced if one takes $G_{\alpha\beta}$ to be the sum of two terms, one of the form (27.14) and the second of the form (27.18). The physical interpretation of adding these two terms together is that some electrons in a solid, such as the valence electrons, are confined to individual atoms and create a magnetic response μ . Another population of electrons creates superconductivity. Adding together these two responses continues to produce a superconductor, but the penetration depth changes and is given by Eq. (27.6).

27.2.2 Thermodynamics of Superconductors

Suppose one takes a long thin cylinder of superconducting material and places it in an external magnetic field of strength H that points along the cylinder axis. The long cylindrical shape is desirable because its demagnetizing factor is zero, and the normal component of the magnetic field automatically vanishes along most of its surface. Experimentally, Meissner and Ochsenfeld (1933) found that at a critical applied field H_c the superconductivity is destroyed and magnetic flux penetrates the sample. The transition is completely reversible, so right at this critical field the superconducting state and normal state of the material must be in equilibrium. The free energy per volume of the metal in the normal state must be just

$$\mathcal{F} = \mathcal{F}_{\text{normal}} + \frac{1}{8\pi\mu} B_c^2. \quad (27.22)$$

Because the superconducting state expels the magnetic induction, its free energy is just $\mathcal{F}_{\text{superconducting}}$.

There now appears to be a problem. The free energy of the normal state is greater than that of the superconducting one, and the magnetic field energy is also positive. Yet when the critical field H_c is applied, (27.22) is supposed to equal $\mathcal{F}_{\text{superconducting}}$. What has gone wrong? The answer is that thermodynamics is being used improperly. The reversible experiments are carried out by slowly varying external currents, not by controlling B directly. In the instant that the normal metal expels B and becomes superconducting, a host of magnetic flux lines blasts off to infinity. The work done during this instant must be accounted for; equivalently, one must choose to work with thermodynamic variables that really do vary smoothly during the change from normal metal to superconductor. The correct thermodynamic potential, as in Eq. (24.27), is

$$\tilde{\mathcal{G}} = \mathcal{F} - \vec{B} \cdot \frac{\delta \mathcal{F}}{\delta \vec{B}} = \mathcal{F}_{\text{normal}} - \frac{1}{8\pi\mu} B_c^2. \quad (27.23)$$

Therefore the difference in free energy per volume at any given temperature between superconducting and normal metal must be exactly

$$\Delta\mathcal{F} \equiv \mathcal{F}_{\text{normal}} - \mathcal{F}_{\text{superconducting}} = \frac{B_c^2}{8\pi\mu}. \quad (27.24)$$

Because $\mu \approx 1$ for superconducting metals, Eq. (27.24) is often written

$$\Delta\mathcal{F} = \frac{H_c^2}{8\pi}. \quad (27.25)$$

One should remember, as mentioned in Section 24.2.3, that the magnetic field H inside a long thin cylinder with axis along the field is always spatially uniform and equal to the externally applied field. It is magnetic induction B that is expelled from superconducting samples, not the magnetic field H .

By differentiating Eq. (27.25) with respect to temperature, one finds immediately that the excess entropy of normal metal relative to the superconductor must be

$$\Delta S = \frac{\partial}{\partial T} \Delta\mathcal{F} = \frac{H_c}{4\pi} \frac{\partial H_c}{\partial T}. \quad (27.26)$$

Meissner and Ochsenfeld determined that as temperature rises towards the critical temperature T_c where superconductivity vanishes in zero magnetic field, the critical field H_c also goes to zero. It follows immediately from Eq. (27.26) that the latent heat of transformation is zero, and therefore the transition is second order.

27.2.3 Landau–Ginzburg Free Energy

The most productive phenomenological description of superconductivity combines ideas from the free energy (27.10) with the idea that the superconductivity resides in a population of superconducting electrons whose density vanishes at a critical temperature T_c . Landau and Ginzburg (1950) described the superconducting state through a macroscopic wave function Ψ that plays a role for superconductivity identical to the role played by the wave function Ψ employed in Eq. (15.98) for superfluid helium. If the supercurrent density $n = |\Psi|^2$ vanishes at T_c , then it should be possible to expand $G_{\alpha\beta}$ (see Eq. (27.10)) in powers of $|\Psi|^2$. Landau and Ginzburg guessed the free energy to be

$$\mathcal{F} = \int \frac{d\vec{r}}{V} \left[\alpha |\Psi|^2 + \frac{\beta}{2} |\Psi|^4 + \frac{1}{8\pi} B^2 + \frac{1}{2m^*} \left| \left[\frac{\hbar}{i} \vec{\nabla} + \frac{2e}{c} \vec{A}(\vec{r}) \right] \Psi(\vec{r}) \right|^2 \right]. \quad (27.27)$$

The first two terms in Eq. (27.27) are procured directly from Landau's general theory of second-order phase transformations, discussed in Section 24.6.1. The remaining terms provide a particular realization of Eq. (27.10) and were inspired by the idea that the superconducting electron wave function might interact with the vector potential like a single macroscopic particle. The effective charge $e^* =$

$2e$ multiplying $\vec{A}$ is due to the fact that pairs of electrons are responsible for the superconductivity. Minimizing Eq. (27.27) with respect to $\vec{A}$ leads to

$$\vec{\nabla} \times \vec{B} = \frac{4\pi}{c} \vec{j}, \quad (27.28)$$

with

$$\vec{j}(\vec{r}) = -\frac{2e\hbar}{2im^*} [\Psi^* \vec{\nabla} \Psi - \Psi \vec{\nabla} \Psi^*] - \frac{4e^2}{m^*c} \vec{A} \Psi^* \Psi. \quad (27.29a)$$

Minimizing with respect to Ψ^* leads to

$$0 = \left[\alpha + \beta |\Psi|^2 + \frac{1}{2m^*} \left(\frac{\hbar}{i} \vec{\nabla} + \frac{2e}{c} \vec{A} \right)^2 \right] \Psi. \quad (27.29b)$$

Integrate so that all spatial derivatives act on Ψ , then take functional derivative with respect to Ψ^* .

Equations (27.29) are the *Landau–Ginzburg equations*.

Boundary Condition. To solve problems in particular geometries, it is always necessary to specify the behavior of Ψ at the boundary of a sample. When a superconductor contacts vacuum no current can flow out of the boundary, so

$$\hat{n} \cdot \left(\frac{\hbar}{i} \vec{\nabla} + \frac{2e}{c} \vec{A} \right) \Psi = 0. \quad (27.30)$$

Here $\hat{n}$ is normal to the boundary; see Landau and Ginzburg (1950), de Gennes (1992), p. 177, or Wertheim (1969), p. 327.

Landau and Ginzburg observe that “it is natural to demand that the wave function at the boundary of the metal should vanish” but such an additional boundary condition on the superconducting wave function would eliminate solutions in thin films except for quantized values of the film thickness. Because this phenomenon is not observed, the seemingly natural boundary condition $\Psi = 0$ must be rejected.

27.2.4 Type I and Type II Superconductors

The Landau–Ginzburg equations depend upon two lengths. The first is a *coherence length* ξ that sets the scale for spatial variations of the order parameter Ψ . This length is given by

$$\xi^2 = \frac{\hbar^2}{2m^*|\alpha|}. \quad (27.31)$$

The second length is the London penetration depth. In terms of Landau and Ginzburg’s parameters it is given by

$$\lambda_L^2 = \frac{m^*c^2\beta}{4\pi|\alpha|(2e)^2}. \quad (27.32)$$

The coherence length and penetration depth, as well as critical temperature and lower critical field are tabulated in Table 27.1.

Table 27.1. Properties of superconductors

Compound	T_c (K)	H_c (G)	ξ (Å)	λ_L (Å)
Al	1.18	105	13 000–16 000	160–500
As ($P = 14$ GPa)	0.5			
Ba ($P = 20$ GPa)	5.3			
Be	0.02			
Bi ($P = 8$ GPa)	8.55			
Cd	0.52	30	7600	1 100
Ce ($P = 5$ GPa)	1.7			
Cs ($P = 13$ GPa)	1.6			
Ga	1.09	58.9		
Hf	0.02			
Hg	3.95	340		380–450
In	3.41	289	2 400–3 500	390–640
Ir	0.10	20.1		
La	6.0	1096		
Lu	0.1			
Mg	0.0005			
Mo	0.92	98		
Nb	9.3	1980	380	390
P ($P = 17$ GPa)	5.8			
Pb	7.20	803	510–960	390–630
Ru	0.49	47		
Se ($P = 13$ GPa)	6.9			
Si ($P = 12$ GPa)	7.1			
Sn	3.7	308	1 000–3 000	340–750
Ta	4.46	831		
Tc	7.8	1410		
Te ($P = 8$ GPa)	4.3			
Th	1.37	162		
Ti	0.42	56		
Tl	2.4	180	4200	
U	1.8			
W	0.02	1.07		
Y ($P = 17$ GPa)	2.7			
Zn	0.85	52		
Zr	0.53	47		
Nb ₃ Sn	18.5	28	34	1 600
YBa ₂ Cu ₃ O _{7-x}	92	500	4–8	900–8 000
HgBa ₂ Ca ₂ Cu ₃ O _y	135			

Critical temperature T_c , lower critical magnetic field H_c where flux first penetrates sample at zero temperature, coherence length ξ and London penetration depth λ_L for selected superconductors. Measurements are at atmospheric pressure unless otherwise indicated. Source: Grigoriev and Meilkhov (1997), pp. 566–567 and Plakida (1995).

Calculation of Coherence Length ξ . These lengths emerge from some simple calculations based upon Eq. (27.29b). First, consider a case in which externally imposed magnetic fields vanish. One sees immediately from Eq. (27.29b) that the spatially uniform solutions are

$$|\Psi|^2 = \begin{cases} \Psi_0^2 \equiv -\frac{\alpha}{\beta} & \text{or} \\ 0. \end{cases} \quad (27.33)$$

The constant β must always be positive or else $\mathcal{F}$ is minimized by sending $\Psi \rightarrow \infty$. The nonzero case in Eq. (27.33) is only possible if $\alpha < 0$, and it corresponds to a spatially uniform superconducting state. The free energy per volume of this state, from Eq. (27.27), is

$$\frac{\mathcal{F}}{\mathcal{V}} = -\frac{\alpha^2}{2\beta} \quad (27.34)$$

and comparing with Eq. (27.25) shows that

$$H_c^2 = \frac{4\pi\alpha^2}{\beta}. \quad (27.35)$$

When superconductivity is present, it makes sense to scale Ψ by this uniform value Ψ_0 and define

$$\psi = \frac{\Psi}{\Psi_0}, \quad (27.36)$$

in terms of which Eq. (27.29b) becomes

$$-\xi^2 \nabla^2 \psi - \psi + \psi|\psi|^2 = 0, \quad \text{When } \vec{A} = 0. \quad (27.37)$$

ξ being given by Eq. (27.31). Thus ξ is the characteristic scale on which ψ varies in the absence of magnetic fields.

One-Dimensional Interface. The physical significance of ξ is illustrated by examining the scale over which the wave function ψ varies. Take ψ to be real, and consider a one-dimensional interface. In one dimension one has

$$-\xi^2 \psi'' - \psi + \psi^3 = 0. \quad (27.38)$$

Multiplying Eq. (27.38) by ψ' and integrating, one has the first integral

$$-\xi^2 (\psi')^2 - \psi^2 + \frac{1}{2} \psi^4 = \text{Const.} \quad (27.39)$$

Equation (27.39) is only consistent with an asymptotic value of $\psi = 1$ if one takes the right hand side constant to be $-1/2$. In this case, one can then write

$$\psi' = \frac{1}{\sqrt{2}\xi} (1 - \psi^2) \quad (27.40)$$

which has a solution

$$\psi = \tanh \frac{x}{\sqrt{2}\xi}. \quad (27.41)$$

This solution describes a transition between two spatially uniform superconducting regions where the phase of the wave function changes sign across the interface. The coherence length ξ sets the spatial scale for this transition.

Penetration Depth. The second length, the penetration depth, arises when one considers a superconductor occupying the region, say, $x < 0$, in an exceedingly weak field. When the field is zero, one has the solution $\Psi = \Psi_0$ for $x < 0$. In the presence of a very weak field, Ψ will change to linear order in the field, but because it is nonzero without a field, to leading order one can continue to use $\Psi = \Psi_0$. Looking at Eq. (27.29a) and keeping only the leading terms gives

$$\vec{j} = \frac{c}{4\pi} \vec{\nabla} \times \vec{B} = -\frac{4e^2}{m^*c} \Psi_0^2 \vec{A}. \quad (27.42)$$

Taking the cross product of Eq. (27.42) gives

$$\vec{\nabla} \times \vec{\nabla} \times \vec{B} = -\frac{4\pi}{c} \frac{4e^2}{m^*c} \Psi_0^2 \vec{B} = -\lambda_L^{-2} \vec{B}. \quad (27.43)$$

Using Eq. (27.33) immediately gives Eq. (27.32) for the penetration depth.

The ratio of the penetration depth and coherence length

$$\kappa = \lambda_L / \xi = \frac{m^*c}{e\hbar} \sqrt{\frac{\beta}{8\pi}} \quad (27.44)$$

is in fact the only free parameter of the Landau–Ginzburg theory, because all other constants can be scaled out, by rescaling ψ , distance, and $\vec{A}$. To be specific, define

$$\vec{a} = \frac{4e\vec{A}}{c\sqrt{2m^*|\alpha|}} \quad (27.45)$$

and measure all distances in units of ξ . Then Eq. (27.29b) becomes

$$\psi - \psi|\psi|^2 - (-i\vec{\nabla} + \vec{a}/2)^2\psi = 0 \quad (27.46)$$

while Eq. (27.29a) becomes

$$\frac{\lambda_L^2}{\xi^2} \vec{\nabla} \times \vec{\nabla} \times \vec{a} = -\frac{1}{i} (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*) - |\psi|^2 \vec{a}. \quad (27.47)$$

Surface Energies. A particularly important result of Landau and Ginzburg's theory is the calculation of the surface energy between normal and superconducting metals. For $\kappa < 1/\sqrt{2}$ the energy is positive; however, if $\kappa > 1/\sqrt{2}$, the surface energy is negative. Once one has a negative surface energy, it becomes favorable for magnetic flux to penetrate a superconductor and form interlocking regions of normal and superconducting metal. Superconductors for which this happens are called *type II*; those for which it does not are called *type I*. The way in which the flux penetrates was worked out by Abrikosov (1957).

In order to discover why the magnitude of κ plays such an important role, consider a superconductor in a magnetic field larger than H_c , so that the superconductivity is destroyed and $\Psi = 0$. Now begin to lower the field, and examine

the linear stability of the state $\Psi = 0$ by expanding everything to first order in Ψ . From Eq. (27.29a), one sees that the current $\vec{j}$ vanishes to this order, and therefore one can take $\vec{A}$ simply to be the vector potential associated with a uniform field H , pointing, say, along the $\hat{z}$ axis. Writing Eq. (27.29b) to linear order then gives

$$\frac{1}{2m^*}(-i\hbar\vec{\nabla} + \frac{2e\vec{A}}{c})^2\Psi = -\alpha\Psi. \quad (27.48)$$

This equation is the same as the eigenvalue equation for an electron in a constant external magnetic field, and so one can immediately write down the solution by comparison with Eq. (25.48). Define H_{c_2} to be the largest magnetic field permitting a nonzero solution of Eq. (27.48). The lowest-energy eigenvalue is $\hbar\omega_c/2$, where in this case

$$\omega_c = \frac{2eH_{c_2}}{m^*c}, \quad (27.49)$$

because in Eq. (27.48) the charge carrier has charge $-2e$. The larger H becomes, the larger α would need to be in order for a solution to be possible. For a given α , the largest H at which Eq. (27.48) has nonzero solution is therefore

$$-\alpha = |\alpha| = \frac{e\hbar H_{c_2}}{m^*c}. \quad (27.50)$$

Using Eq. (27.35) to express the critical field H_c in terms of α and β , along with Eq. (27.44) to define κ , gives

$$\frac{H_{c_2}}{H_c} = \sqrt{2\kappa}. \quad (27.51)$$

Now one can see why $\kappa = 1/\sqrt{2}$ divides the two types of superconductors. If $\kappa > 1/\sqrt{2}$, then H_{c_2} lies above H_c . For applied magnetic fields between H_c and H_{c_2} , it is energetically unfavorable to expel all magnetic flux from the system, but favorable to form at least one superconducting vortex, corresponding to solutions of Eq. (27.48). Equation (27.48) predicts that a vortex will form, but is unable to describe its strength or final form; these can only be obtained from more elaborate calculations taking into account the nonlinear terms of Eq. (27.47). As the magnetic field descends below H_{c_2} toward H_c , more and more vortices fill the system, until at H_c they coalesce and expel all the magnetic flux. By contrast, if $\kappa < 1/\sqrt{2}$, forming a vortex does not become possible until the magnetic field has dropped below H_c . It is reasonable to guess, and possible to verify, that the system does not choose to form vortices when it has the option of expelling all the magnetic field and to become superconducting instead. The picture that led to Eq. (27.48), in which a normal metal in a magnetic field develops a superconducting wave function to order Ψ , is no longer valid. It is more favorable for Ψ to grow to size $-\alpha/\beta$ and take over the whole sample.

Interface Energies. Problem 2 describes the free energies of superconducting-metal interfaces in two limits. For $\kappa = 0$,

$$\frac{\tilde{g}}{A} = \frac{H_c^2}{4\pi} \sqrt{2} \xi \frac{2}{3}, \quad (27.52)$$

and forming the interface costs a finite positive energy per unit area A . For $\kappa \rightarrow \infty$,

$$\frac{\tilde{G}}{A} = -\frac{H_c^2}{8\pi}\lambda_L; \quad (27.53)$$

because this energy is negative, the system will respond by trying to form as much interface as it can. The result is the *Abrikosov lattice* of flux lines penetrating the sample depicted in Figure 27.2.

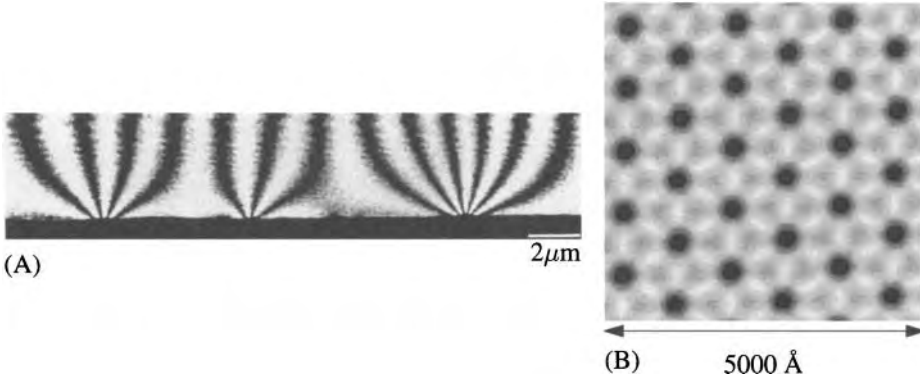


Figure 27.2. A Type II superconductor is unstable to the formation of flux tubes that penetrate the sample trying to generate a maximal area where superconductor and metal are in contact. The lowest free energy configuration is often a triangular lattice. (A) Electron hologram from the side of magnetic flux entering a lead film [Source: Tonomura et al. (1986), p. 93.] (B) Top view of an Abrikosov lattice of flux tubes in NbSe₂, taken with scanning tunneling microscopy. [Courtesy of S. Pan and A. de Lozanne, University of Texas.]

27.2.5 Flux Quantization

The Landau–Ginzburg equations have an effective charge $-2e$ seated in the spot one would expect to be occupied by the electron charge $-e$. This effective charge has a convoluted history. Ginzburg had guessed that the superconducting charge carriers might have an effective charge e^* . Landau objected that effective charges violate gauge invariance, and their paper takes the position that “ e is a charge, which there is no reason to consider as different from the electronic charge.” In fact, arguments based upon gauge invariance require only that charges be integral multiples of the electron charge; in two-dimensional space these arguments permit effective particles with arbitrary charge, as shown in Figure 25.11 and discussed by Wilczek (1990). Despite the fact that the microscopic theory of superconductivity involved pairing of electrons, the correct form of the Landau–Ginzburg equations was not clear, and the effective charge $-2e$ was a surprise when discovered experimentally by Deaver and Fairbank (1961) and Doll and Näbauer (1961) in studies of quantized magnetic fluxes.

The Landau–Ginzburg equations predict that the magnetic flux trapped within a superconductor is quantized (Figure 27.3). Suppose, however, that the effective

charge $-e^*$ and mass m^* of the superconducting particles were not known. The superconducting current associated with Ψ would be

$$\vec{j} = -\frac{e^*\hbar}{2im^*}[\Psi^*\vec{\nabla}\Psi - \Psi\vec{\nabla}\Psi^*] - \frac{e^{*2}}{m^*c}\vec{A}\Psi^*\Psi. \quad (27.54)$$

Letting

$$\Psi(\vec{r}) = \Psi_0 e^{i\phi(\vec{r})} \quad \text{Take } \phi \text{ to be real.} \quad (27.55)$$

gives

$$\vec{j} = -\frac{\Psi_0^2}{m^*} \left(\frac{e^{*2}}{c}\vec{A} + e^*\hbar\vec{\nabla}\phi \right) \quad (27.56)$$

$$\Rightarrow -\vec{\nabla}\phi = \frac{1}{\hbar} \left(\frac{m^*}{e^*\Psi_0^2}\vec{j} + \frac{e^*}{c}\vec{A} \right). \quad (27.57)$$

Simply demanding that the phase ϕ be differentiable is enough to prove that the magnetic flux is quantized, just as in Eq. (15.106). Because Ψ must return to itself when ϕ changes through 2π , one obtains

$$-\int d\vec{s} \cdot \vec{\nabla}\phi = 2\pi l, \quad l \text{ is an integer.} \quad (27.58)$$

As a result,

$$\int d\vec{s} \cdot \frac{1}{\hbar} \left[\frac{m^*}{e^*\Psi_0^2}\vec{j} + \frac{e^*}{c}\vec{A} \right] = 2\pi l. \quad (27.59)$$

Choosing a contour of integration that lies within the cylinder at a depth greater than the skin depth (Figure 27.3), $\vec{j}$ is negligible, and

$$\frac{e^*}{ch} \int d\vec{s} \cdot \vec{A} = 2\pi l \quad (27.60)$$

$$\Rightarrow \int d^2r B_z = \Phi = \frac{2\pi\hbar c}{e^*} = l \frac{e}{e^*} \Phi_0. \quad \text{See definition of } \Phi_0 \text{ in Eq. (25.51).} \quad (27.61)$$

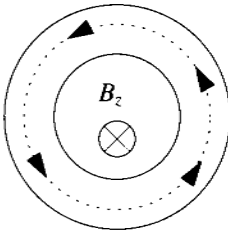


Figure 27.3. Magnetic flux that pierces a superconducting ring is quantized in units of $\Phi_0/2$. The integration contour in Eq. (27.59) is taken on the dotted line, which lies deep in the superconductor where $\vec{j}$ vanishes.

So the integrated magnetic flux Φ that penetrates a hole through a superconductor is quantized in units of $(e/e^*)\Phi_0$, and can be used to measure the effective charge e^* . The measurements in Figure 27.4 show that $e^* = 2e$, leading to the conventional form of Eq. (27.27). The fact that the only phase changes observed in Figure 25.5 are 0 and π is another experimental demonstration of this same fact.

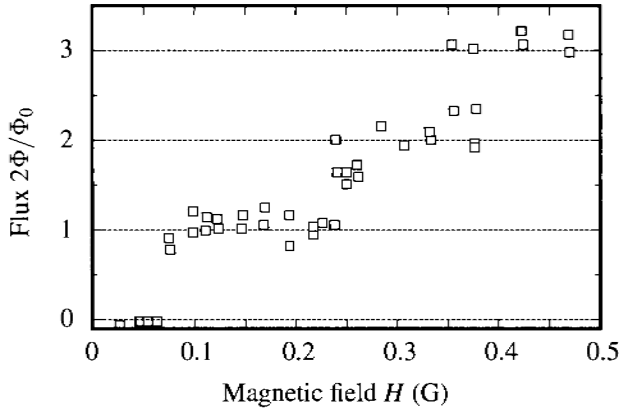


Figure 27.4. Trapped magnetic flux in a superconducting cylinder as a function of applied field. The dimensions of the cylinder are on the order of $10 \mu\text{m}$, and it is made of tin. The flux is measured by moving the cylinder up and down at 100 Hz toward and away from a conducting coil and measuring the electromotive force induced in the coil. [Source: Deaver and Fairbank (1961), p. 44; a nearly identical experiment was published simultaneously by Doll and Näbauer (1961). Theoretical discussions by Byers and Yang and Onsager are sandwiched between these two experimental papers.]

27.2.6 The Josephson Effect

It takes a certain degree of courage to adopt a phenomenological function like Ψ , treat it as a physical entity, and predict new phenomena. Josephson (1962) predicted that the wave function could be induced to interfere with itself and oscillate if two superconductors were separated by a small strip of nonsuperconducting material. A qualitative derivation of Josephson's equations is presented at first; Section 27.2.9 obtains the basic results in a more careful way.

Consider two superconductors separated by a thin insulating layer. On the opposite sides of this *weak link*, the two superconductors interact only because of their exponentially small residues as they decay across the barrier into each others territory. The change in energy of two nearly independent systems due to tunneling is given in general by

$$\int d\vec{r} U(\vec{r}) (\Psi_1^*(\vec{r}) \Psi_2(\vec{r}) + \Psi_1(\vec{r}) \Psi_2^*(\vec{r})) \quad (27.62)$$

$$= \epsilon (\Psi_1^* \Psi_2 + \Psi_1 \Psi_2^*) , \quad (27.63)$$

where the wave function without the argument $\vec{r}$ refers to its value at some reference point inside the bulk, and ϵ is some very small quantity with dimensions of energy. Writing down a Schrödinger equation for two wave functions coupled by such an interaction leads to

$$\frac{\partial \Psi_1}{\partial t} = \frac{-i}{\hbar} [\mathcal{E}_1 \Psi_1 + \epsilon \Psi_2] \quad (27.64a)$$

$$\frac{\partial \Psi_2}{\partial t} = \frac{-i}{\hbar} [\mathcal{E}_2 \Psi_2 + \epsilon \Psi_1] . \quad (27.64b)$$

Taking each wave function now to be of the form

$$\Psi_l = \sqrt{n_l} e^{i\phi_l} \quad (27.65)$$

gives

$$\left(\frac{1}{2} \frac{\dot{n}_1}{\sqrt{n_1}} + i\sqrt{n_1} \dot{\phi}_1 \right) e^{i\phi_1} = \frac{-i}{\hbar} \left[\mathcal{E}_1 \sqrt{n_1} e^{i\phi_1} + \epsilon \sqrt{n_2} e^{i\phi_2} \right]. \quad (27.66)$$

A similar equation holds with indices reversed.

The superconducting electron densities n_1 and n_2 within the superconducting regions are the same and constant, apart from some extremely small variations, so by taking the real and imaginary parts of Eq. (27.66), one finds

$$\dot{n}_1 = 2 \frac{en}{\hbar} \sin(\phi_2 - \phi_1) = -\dot{n}_2 = \frac{j}{2e} \quad (27.67a)$$

Superconducting particles are flowing from one superconductor to another; this time rate of change is the total current.

$$\dot{\phi}_2 - \dot{\phi}_1 = \frac{1}{\hbar} (\mathcal{E}_1 - \mathcal{E}_2) = 2e(V_2 - V_1)/\hbar. \quad (27.67b)$$

The factor of two appears at the end because the energies correspond to energies of electron pairs.

In the presence of magnetic fields, Eqs. (27.67) are not correct because the phase of a wave function changes in the presence of a magnetic field. To make a wave function phase gauge-invariant, add $2e/(\hbar c) \int_0 d\vec{s} \cdot \vec{A}$ to the phase, where the line integral is taken from some arbitrary reference point. Incorporating this generalization, Josephson's equations become

$$\vec{j} = \vec{j}_0 \sin(\phi_2 - \phi_1 + \frac{2e}{\hbar c} \int_1^2 d\vec{s} \cdot \vec{A}) \quad (27.68a)$$

$$\frac{-1}{\hbar} (\mathcal{E}_2 - \mathcal{E}_1) = 2eV/\hbar = \frac{\partial}{\partial t} \left(\phi_2 - \phi_1 + \frac{2e}{\hbar c} \int_1^2 d\vec{s} \cdot \vec{A} \right). \quad (27.68b)$$

V is the voltage difference between the two superconductors, and the factor of two appears because the energies $\mathcal{E}_l$ are energies of electron pairs.

Equations (27.68) have peculiar properties. If one places two superconductors at different voltages in contact ($\mathcal{E}_1 - \mathcal{E}_2 \neq 0$), then $\phi_2 - \phi_1$ drifts in time, so n_1 and n_2 oscillate about their mean values at $484 \text{ MHz}/\mu\text{V}$ —the *AC Josephson effect*. On the other hand, if $\mathcal{E}_1 - \mathcal{E}_2 = 0$, then $\phi_2 - \phi_1$ does not change in time, but does not have to vanish, so Eq. (27.68a) permits steady current flow—the *DC Josephson effect*. If a current source injects electrons into a Josephson junction, then the phase difference $\phi_2 - \phi_1$ simply adjusts to a nonzero value, up to a maximum value of $\pi/2$. Current flow in the absence of a voltage difference is a defining properties of superconductors.

The experiment that was decisive in bringing about acceptance of the Josephson effect was performed by Anderson and Rowell (1963). It consisted of measuring the relationship between current and voltage in a *Josephson junction* as a function of magnetic field. Later experiments probed more exotic interference effects; those shown in Figure 27.5 provide a clear demonstration that Eqs. (27.68) are sound. Calculation of the form of the interference is left to Problem 3.

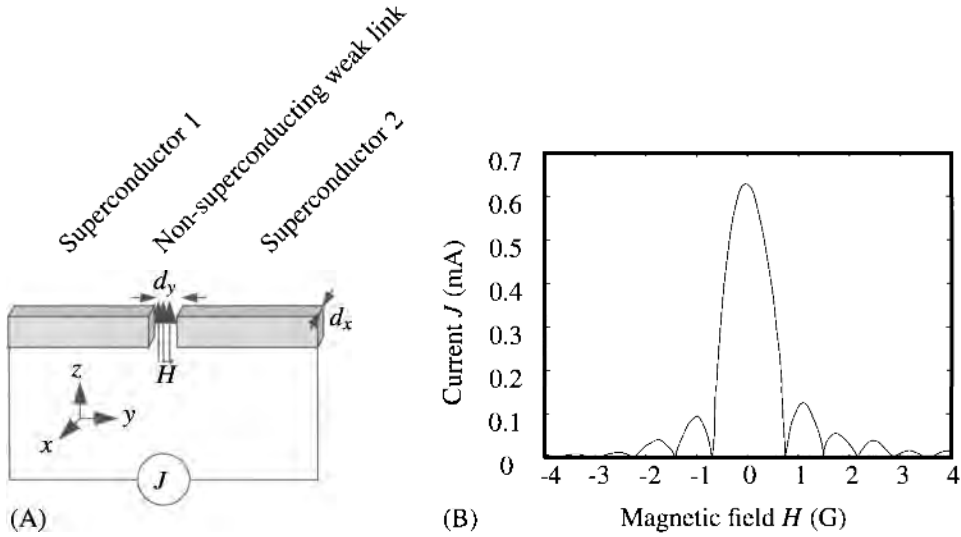


Figure 27.5. (A) Setting for Fraunhofer diffraction in a Josephson junction. (B) The critical Josephson current J_c is the maximum zero-voltage current that can flow through a Josephson junction. The data show a measurement of J_c in an Sn–SnO–Sn junction at $T = 1.9$ K. [Results of R. C Jaklevic, reported by Mercereau (1969), p. 402]

27.2.7 Circuits with Josephson Junction Elements

When used as a circuit element, the Josephson junction has both some resistance R and capacitance C . So the circuit element containing the Josephson junction is described by

$$\frac{V}{R} + J_0 \sin \phi + C\dot{V} = J, \quad \text{Leave out magnetic fields for the moment so as not to have to carry along the integrals of the vector potential.} \quad (27.69)$$

where $\phi = \phi_2 - \phi_1$. Eq. (27.69) constitutes the *resistively shunted junction* (RSJ) model. In the absence of magnetic fields

$$\dot{\phi} = 2eV/\hbar, \quad (27.70)$$

because the difference in Fermi energies between the two superconductors is just $2eV$. Therefore

$$J = \frac{\dot{\phi}\hbar}{2eR} + J_0 \sin \phi + \frac{C\hbar}{2e} \ddot{\phi} \quad (27.71)$$

$$\Rightarrow \frac{\hbar C}{2e} \ddot{\phi} + \frac{\hbar}{2eR} \dot{\phi} = -\frac{\partial}{\partial \phi} [-\phi J - J_0 \cos \phi]. \quad (27.72)$$

Equation (27.72) has a mechanical analog, the motion of a damped particle in a potential. The potential slopes downwards, but with periodic bumps as shown

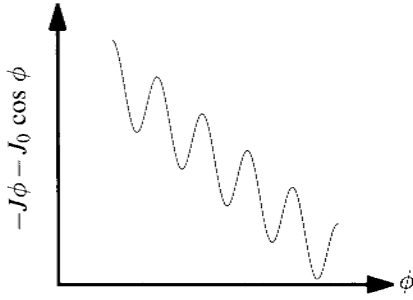


Figure 27.6. The washboard potential in Eq. (27.72).

in Figure 27.6: It is called a *tilted washboard potential*. One can understand the behavior of the Josephson junction based on the mechanical analogy without going through complicated analysis. First, make the equations dimensionless by measuring time in units of

$$t_0 = \frac{\hbar}{2eJ_0R}, \quad (27.73)$$

so that Eq. (27.72) becomes

$$\beta \ddot{\phi} + \dot{\phi} = -\frac{\partial}{\partial \phi} \left[-\phi \frac{J}{J_0} - \cos \phi \right], \quad (27.74)$$

where

$$\beta = \frac{J_0 R^2 C 2e}{\hbar}. \quad (27.75)$$

If $\beta \gg 1$, then the particle is very heavy, and the junction is hysteretic. To see why, suppose the current J to be less than J_0 . The particle sits in one of the basins, perhaps rocking back and forth slightly, meaning that the voltage oscillates weakly about zero. Next suppose the current J to be raised above J_0 . Then the particle rolls down the washboard. This continual motion corresponds to a voltage V . If the current is lowered again to its original value, the ball has acquired momentum, and because the damping is weak, it will continue to roll across the washboard. The nonzero voltage maintains itself, although the current is less than critical. This hysteretic behavior is prevented when the particle is overdamped, which corresponds to a small shunting resistance R .

27.2.8 SQUIDS

Superconducting quantum interference devices (SQUIDS) are Josephson junctions employed as very sensitive detectors of magnetic flux. DC SQUIDS consist of superconducting loops containing two junctions. Consider the contour depicted in Figure 27.7. Assuming that the superconductors are thicker than the penetration depth, along with Eq. (27.57), one obtains $\vec{\nabla} \phi = -4\pi \vec{A} / \Phi_0$ everywhere in the superconductor. Therefore

$$\oint d\vec{s} \cdot \vec{A} = \Phi = \int_4^1 d\vec{s} \cdot \vec{A} - \frac{\Phi_0}{4\pi} \int_1^2 d\vec{s} \cdot \vec{\nabla} \phi + \int_2^3 d\vec{s} \cdot \vec{A} - \frac{\Phi_0}{4\pi} \int_3^4 d\vec{s} \cdot \vec{\nabla} \phi \quad (27.76)$$

$$\Rightarrow \Phi = \frac{\Phi_0}{4\pi} (\gamma_{23} - \gamma_{14}), \quad (27.77)$$

where

$$\gamma_{14} = \phi_4 - \phi_1 + \frac{4\pi}{\Phi_0} \int_1^4 d\vec{s} \cdot \vec{A}. \quad \text{With an identical expression for } \gamma_{23}. \quad (27.78)$$

If the weak links were absent, the phase changes γ_{14} and γ_{23} would have to be multiples of 2π , giving back the flux quantization condition (27.61). The gaps between the superconductors allow arbitrary amounts of flux to creep into the loop, but then place a constraint on the phases of the wave function.

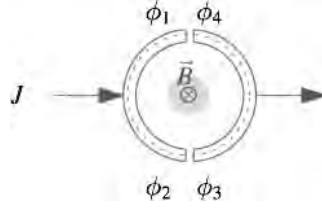


Figure 27.7. A DC SQUID consists of two Josephson junctions in parallel. The assembly is extremely sensitive to the enclosed magnetic flux. The dotted line is a contour used to calculate Eq. (27.76).

According to Eq. (27.68a), the total current passing through the DC SQUID is

$$J = J_0 \sin(\gamma_{14}) + J_0 \sin(\gamma_{23}) \quad (27.79)$$

$$= J_0 \left[\sin(\gamma_{23} - 4\pi\Phi/\Phi_0) + \sin(\gamma_{23}) \right]. \quad \text{Inserting Eq. (27.77).} \quad (27.80)$$

The current oscillates as a function of the phase Φ in the SQUID, and it can be used as a sensitive measure of magnetic fields. In practice, Eq. (27.69) must be used to describe the behavior of the phase, rather than Eq. (27.68a), and the junctions are resistively shunted so as to eliminate the hysteresis that would accompany the completely nondissipative case.

27.2.9 Origin of Josephson's Equations

Section 27.2.1 described the derivation of superconductivity from hypotheses concerning a free energy. A broadening of this point of view to encompass dynamical phenomena allows for a very general derivation of Josephson's equations.

The new point of view focuses upon the field ϕ appearing in the macroscopic wave function $\Psi_0 \exp i\phi$. According to Eq. (27.57), so long as Ψ_0 is constant, the current $\vec{j}$ can be put in the form

$$\vec{j} = -\frac{|\Psi_0|^2 8\pi e\hbar}{m^* \Phi_0} \left[\frac{\Phi_0}{4\pi} \vec{\nabla} \phi + \vec{A} \right]. \quad \text{The magnetic flux quantum } \Phi_0 \text{ is defined in Eq. (25.51).} \quad (27.81)$$

Therefore $\chi = \Phi_0 \phi / 4\pi$ enters the theory in exactly the same fashion as the field χ in electrodynamics used to generate gauge transformations of the vector potential.

The most economical theory of superconductivity begins with two postulates:

1. The field χ generating phase changes has become a macroscopic field with physically measurable consequences. The vector potential always appears in combination with χ as $\vec{A} + \vec{\nabla}\chi$, while the scalar potential V always appears in combination with χ as $V - \dot{\chi}/c$.
2. When $\phi = 4\pi\chi/\Phi_0$ appears in the theory, it does so in the form $\exp[i\phi]$, so ϕ only has significance up to multiples of 2π .

These postulates describe a *spontaneous breaking of gauge symmetry*.

Having determined that the gauge potential χ should be part of the theory, one can write down a Lagrangian in the spirit of Section B.3 for superconductivity in the form

$$L = \int d\vec{r} dt \mathcal{L} = \int d\vec{r} dt \left\{ \frac{E^2 - B^2}{8\pi} - G(\vec{A} + \vec{\nabla}\chi, V - \dot{\chi}/c) \right\}. \quad V \text{ is the scalar potential.} \quad (27.82)$$

Use

$$\vec{E} = -\vec{\nabla}V - \frac{1}{c} \frac{\partial \vec{A}}{\partial t}, \quad \text{and} \quad \vec{B} = \vec{\nabla} \times \vec{A}. \quad (27.83)$$

Problem 4 demonstrates through straightforward computations that

$$\frac{\delta L}{\delta V} = 0 \Rightarrow \frac{\partial G}{\partial V} = -ne \quad -ne \text{ is the charge density.} \quad (27.84a)$$

$$\frac{\delta L}{\delta \vec{A}} = 0 \Rightarrow \frac{\partial G}{\partial \vec{A}} = -\frac{\vec{j}}{c}. \quad (27.84b)$$

The Lagrangian must also vanish with respect to variations in χ , so

$$\frac{\delta L}{\delta \chi} = 0 \Rightarrow \vec{\nabla} \cdot \frac{\partial G}{\partial \vec{A}} - \frac{\partial}{\partial t} \frac{1}{c} \frac{\partial G}{\partial V} = 0 \quad (27.85)$$

$$\Rightarrow \frac{\partial}{\partial t} [-ne] = -\vec{\nabla} \cdot \vec{j}. \quad \text{This is just the equation of continuity, obtained by substituting in Eqs. (27.84).} \quad (27.86)$$

The Lagrangian depends upon the fields $\vec{A}$, V , and χ , as well as upon the time derivatives of $\vec{A}$ and χ . Therefore, the Hamiltonian is

$$\mathcal{H} = \vec{A} \cdot \frac{\partial \mathcal{L}}{\partial \vec{A}} + \dot{\chi} \frac{\partial \mathcal{L}}{\partial \dot{\chi}} - \mathcal{L}. \quad \text{Just the definition of the Hamiltonian, as on p. 346 of Goldstein (1980).} \quad (27.87)$$

There is no need to calculate any properties of the Hamiltonian explicitly in order to derive the Josephson effect. One has only to observe that the Hamiltonian is conserved and must equal the energy of the system. According to Eq. (27.84a), the canonical momentum corresponding to χ is proportional to the charge density,

$$\frac{\partial \mathcal{L}}{\partial \dot{\chi}} = -\frac{ne}{c}. \quad (27.88)$$

Therefore Hamilton's equations for χ must be

$$\frac{\partial \mathcal{H}}{\partial \chi} = \frac{\dot{n}e}{c} \quad \text{Identify } Q \text{ with } \chi \text{ and } P \text{ with } -ne/c. \text{ This equation is } \frac{\partial \mathcal{H}}{\partial Q} = -\dot{P}. \quad (27.89a)$$

$$\frac{\partial \mathcal{H}}{\partial [-ne/c]} = \dot{\chi}. \quad \text{This equation is } \frac{\partial \mathcal{H}}{\partial P} = \dot{Q}. \quad (27.89b)$$

The derivative of the energy $\mathcal{H}$ with respect to the electron density n is precisely the electron electrochemical potential μ . Finally,

$$\dot{\chi} = -\frac{c\mu}{e} \Rightarrow \dot{\phi} = -\frac{2\mu}{\hbar} = \frac{2eV}{\hbar}. \quad \text{Use } \chi = \phi\hbar c/2e. \quad (27.90)$$

If two points differ in voltage by an amount $\delta V = \delta\mu/(-e)$, then the phase difference $\delta\phi$ between these points changes at a rate $\delta\dot{\phi} = 2e\delta V/\hbar$. Because all physical quantities are periodic functions of $\phi + 4\pi \int d\vec{s} \cdot \vec{A}/\Phi_0$ with period 2π , the current must oscillate with frequency $2e\delta V/\hbar$. The arguments leading to this conclusion depend only upon the assumption that ϕ is a physical field and that physical quantities depend upon $\exp[i\phi]$. Oscillations of Josephson junctions therefore provide the most precise measurement of the *Josephson constant* $K_J = 2e/\hbar$.

While Eq. (27.68b) depends only upon fundamental constants, and is correct to very high accuracy, Eq. (27.68a) is less fundamental. The function $\sin \phi$ appearing in Eq. (27.68a) could be replaced by other functions, just so long as they had the same periodicity.

27.3 Microscopic Theory of Superconductivity

The phenomenological theories of superconductivity are in many respects quite complete. Yet there are certain types of questions that they cannot answer. Why are some materials superconducting and others not? What determines the critical temperature T_c and critical field H_c ? What sets the coherence length and penetration depth λ_L ? These are issues that a microscopic theory should address. Because the single-electron Hamiltonians that are amenable to exact or numerical solution do not produce superconductivity and because the full many-electron Hamiltonian is intractable, decades passed after the experimental discovery of superconductivity with little theoretical progress:

Bloch, in a famous theorem later extended by Bohm to many-body systems, showed that in the absence of a magnetic field the most stable state of an electron system is that of zero current. Because of the frustrations which the many theorists who worked on the problem encountered, Bloch jokingly proposed a second theorem—that “any theory of superconductivity can be refuted.”
—Bardeen (1963), p. 3

A crucial step on the path to a microscopic theory was the discovery of the isotope effect by Maxwell (1950) and Reynolds et al. (1950), shown in Figure 27.8.

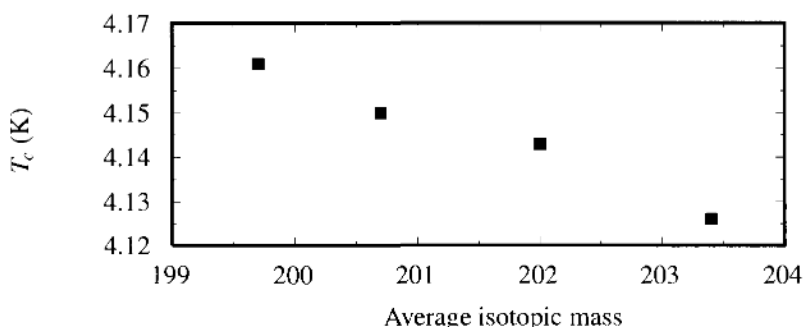


Figure 27.8. Superconducting transition temperature T_c versus average isotopic mass in four samples of mercury, showing decrease in T_c with increasing mass. [Source: Reynolds et al. (1950), p. 487.]

If the superconducting transition temperature could be depressed by changing the mass of the ions, then the interaction of electrons and ions must be crucial. The theory of polarons in Section 22.3.2 is based upon this same type of interaction. In fact, Fröhlich (1950) began a theory of superconductivity based upon an interaction between electrons and phonons prior to the experimental observations shown in Figure 27.8, and the Hamiltonian he deduced for this purpose is a starting point for successful microscopic models.

27.3.1 Electron–Ion Interaction

Fröhlich (1950) found that electrons could effectively attract each other because of an intermediate interaction with phonons. In rough terms, the motion of an electron with wave number $\vec{k}$ through the lattice causes the ions to vibrate with the same wave number. It is then energetically favorable for a second electron, traveling in the opposite direction from the first but with the same wavelength, to synchronize its phase with the first electron so as to obtain the greatest energetic benefit from the ionic vibrations. This synchrony is an effective attraction between the two electrons. To see in detail how the attraction comes about requires bringing together results on the interaction of electrons with ions and with each other.

The most physically meaningful way to perform this task is by imagining that an electric field $\vec{E}$ has been imposed on a metal and then finding the resulting motion of charge, including both (a) charge motion associated with conduction electrons and (b) charge motion associated with ions. From the total conductivity σ obtained in this way follows a dielectric function ϵ that contains information on all the charge carriers bundled together. This dielectric function shows how electrons and ions adjust in response to any electric field, but in particular it can be used to show how electrons and ions adjust in response to the electric field of a moving electron. It follows that the effective interaction of two electrons separated by distance r is $e^2/\epsilon r$, with ϵ carrying information about the cooperative motion both of ions and a sea of conduction electrons.

Conduction Electron Current. Equations (20.75) and (20.76) describe how conduction electrons move in response to external potentials V . Because $\sigma = \omega(\epsilon - 1)/4\pi i$ from (20.14), Eq. (20.84) says that for longitudinal waves, the conduction electron contribution to the conductivity is

$$\sigma_{\text{el}} = \frac{i\omega\chi_c}{q^2}. \quad (27.91)$$

Equation (27.91) looks simple until one contemplates writing χ_c out in its full splendor. Evaluation of Eq. (20.76) can be simplified in two steps:

1. Only the low-frequency susceptibility is needed. The reason is that electrons will only experience an attractive interaction when they oscillate at frequencies characteristic of phonons. As shown in Figure 20.1, characteristic phonon energies and frequencies are around two orders of magnitude less than characteristic electron energies and frequencies, so χ_c can be evaluated at $\omega = 0$, allowing the electrons to respond adiabatically to the phonons.
2. Only the long wavelength susceptibility is needed. When $\omega \rightarrow 0$, Problem 20.6 shows that χ_c takes the form

$$\chi_c = -\frac{me^2}{\pi^2\hbar^2} \frac{(4k_F^2 - q^2) \log\left(\frac{q + 2k_F}{2k_F - q}\right) + 4k_F q}{8q} \quad (27.92)$$

and that in the low q limit

$$\chi_c = -\frac{me^2 k_F}{\pi^2\hbar^2} \equiv -\frac{\kappa_c^2}{4\pi}. \quad (27.93)$$

There seems not to be much justification for going to the low q limit because the interest will be in q in the neighborhood of k_F . However, the plot of $\chi_c(q)/\chi_c(0)$ in Figure 27.9 shows that the error is hardly significant, because χ changes only by 10% by the time q reaches k_F .

The final result is that the contribution of electrons to the conductivity is

$$\sigma_{\text{el}} = \frac{\omega\kappa_c^2}{4\pi i q^2}. \quad (27.94)$$

Ion Current. Section 22.3.2 showed how an electron moving in a polar solid would cause ions to move. The analysis relevant for superconductivity is very similar, but it must be modified slightly because the important phonons are acoustic rather than optical.

Returning to Eq. (22.18), suppose that ionic displacements $\vec{u}$ respond to applied electric fields by obeying

$$\vec{u} = \frac{-e^* \vec{E}}{M(\omega^2 - \bar{\omega}_q^2)}. \quad (27.95)$$

For small $\vec{q}$, $\vec{E}$ should probably be replaced by $\vec{E}_{\text{cell}}$, but in view of other uncertainties the replacement is pedantic.

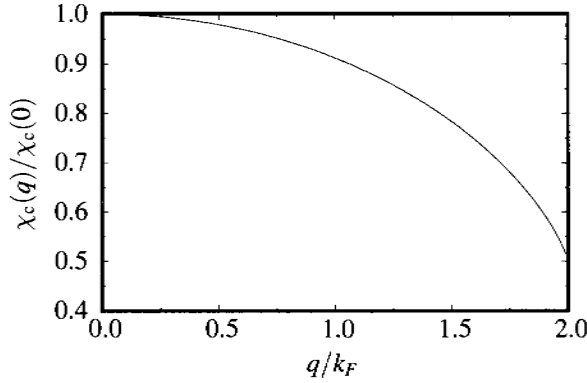


Figure 27.9. Charge susceptibility χ_c , showing that the large wavelength limit $q \rightarrow 0$ is a satisfactory approximation for q as large as k_F .

In Section 22.3.2 the bare phonon frequency $\bar{\omega}$ was constant; now take it to depend upon $\vec{q}$, and in particular to vanish for $\vec{q} = 0$. The function $\bar{\omega}_{\vec{q}}$ is assumed to be known, but its particular form will not be important in what follows here. Observing that

$$\vec{f}_{\text{ion}}(\vec{q}, \omega) = -i\omega n e^* \vec{u} \quad (27.96)$$

and defining

$$\omega_{\text{pi}}^2 = \frac{4\pi n e^{*2}}{M}, \quad \text{This is an ionic plasma frequency.} \quad (27.97)$$

the conductivity σ_{ion} of the ions is

$$\sigma_{\text{ion}} = -\frac{\omega}{4\pi i} \frac{\omega_{\text{pi}}^2}{\omega^2 - \bar{\omega}_{\vec{q}}^2}. \quad (27.98)$$

Electrons and Ions Together. When one finally combines electronic and ionic currents, the total conductivity is

$$\sigma(\vec{q}, \omega) = \frac{\omega}{4\pi i} \left[\frac{\kappa_c^2}{q^2} - \frac{\omega_{\text{pi}}^2}{\omega^2 - \bar{\omega}_{\vec{q}}^2} \right] \quad \text{Combine Eqs. (27.98) and (27.94).} \quad (27.99)$$

$$\Rightarrow \epsilon(\vec{q}, \omega) = 1 + \frac{\kappa_c^2}{q^2} - \frac{\omega_{\text{pi}}^2}{\omega^2 - \bar{\omega}_{\vec{q}}^2}. \quad \text{See Eq. (20.14).} \quad (27.100)$$

Just as in polar solids, $\bar{\omega}_{\vec{q}}$ does not correspond to an actual resonant frequency. The frequency $\omega_{\vec{q}}$ of longitudinal phonons is given by demanding that ϵ vanish, and it is

$$\omega_{\vec{q}}^2 = \bar{\omega}_{\vec{q}}^2 + \frac{q^2 \omega_{\text{pi}}^2}{q^2 + \kappa_c^2}. \quad \text{See Eq. (20.23). Notice that if electrons were not available to screen the ions, the longitudinal phonons would oscillate at the plasma frequency } \omega_{\text{pi}} \text{ as } \vec{q} \rightarrow 0, \text{ rather than producing the expected acoustic mode.} \quad (27.101)$$

Using this definition, a bit of algebra shows that

$$\frac{1}{\epsilon(\vec{q}, \omega)} = \frac{q^2}{q^2 + \kappa_c^2} \left[\frac{\omega^2 - \bar{\omega}_{\vec{q}}^2}{\omega^2 - \omega_{\vec{q}}^2} \right]. \quad (27.102)$$

Effective Interaction. Suppose now that two electrons move through a solid whose dielectric function is given by Eq. (27.102). If they have wave vectors $\vec{k}_1$ and $\vec{k}_2$ and energies $\mathcal{E}_1$ and $\mathcal{E}_2$, then the charge density of the two of them oscillates as

$$|\psi_1 e^{i\vec{k}_1 \cdot \vec{r} - \mathcal{E}_1 t/\hbar} + \psi_2 e^{i\vec{k}_2 \cdot \vec{r} - \mathcal{E}_2 t/\hbar}|^2 \propto \text{const.} + \cos[(\vec{k}_1 - \vec{k}_2) \cdot \vec{r} - (\mathcal{E}_1 - \mathcal{E}_2)t/\hbar]. \quad (27.103)$$

Their effective interaction is therefore

$$U_{\text{eff}} = \frac{4\pi e^2}{\epsilon(\vec{q}, \omega) q^2} = \frac{4\pi e^2}{q^2 + \kappa_c^2} \left[1 + \frac{\omega_{\vec{q}}^2 - \bar{\omega}_{\vec{q}}^2}{\omega^2 - \omega_{\vec{q}}^2} \right] \quad (27.104a)$$

with

$$\vec{q} = \vec{k}_1 - \vec{k}_2 \quad \text{and} \quad \hbar\omega = \mathcal{E}_1 - \mathcal{E}_2. \quad (27.104b)$$

The essential point to observe in Eqs. (27.104) is that when $\bar{\omega}_{\vec{q}} < \omega < \omega_{\vec{q}}$, the effective interaction becomes negative, and the two electrons attract one another.

Second Quantized Interaction.

To see how to express the interaction of electrons in second quantized notation, suppose that electrons have an effective potential $U_{\text{eff}}(\vec{r} - \vec{r}')$ in position space. Use plane waves $\exp[i\vec{k} \cdot \vec{r}]/\sqrt{V}$ as the basis states in Eq. (C.10). Then the interaction becomes

$$\hat{U}_{\text{eff}} = \frac{1}{2} \sum_{\vec{k}\vec{k}'\vec{k}''\vec{k}'''} \hat{c}_{\vec{k}}^\dagger \hat{c}_{\vec{k}'}^\dagger \hat{c}_{\vec{k}''} \hat{c}_{\vec{k}'''} U_{\text{eff}}(\vec{r}_1 - \vec{r}_2) |\vec{k}''\vec{k}'''\rangle \quad (27.105)$$

The matrix element becomes

$$\frac{1}{V^2} \int d\vec{r} d\vec{r}' d\vec{r}'' d\vec{r}''' \left[e^{-i\vec{k} \cdot \vec{r} - i\vec{k}' \cdot \vec{r}' + i\vec{k}'' \cdot \vec{r}'' + i\vec{k}''' \cdot \vec{r}'''} U_{\text{eff}}(\vec{r} - \vec{r}') \right. \\ \left. \times \delta(\vec{r} - \vec{r}'') \delta(\vec{r}'' - \vec{r}''') \right] \quad (27.106)$$

$$= \frac{(2\pi)^3}{V^2} \delta(\vec{k}'' + \vec{k}''' - \vec{k} - \vec{k}') U_{\text{eff}}(\vec{k}'' - \vec{k}) \quad (27.107)$$

$$= \frac{1}{V} \delta_{\vec{k}'' + \vec{k}''' - \vec{k} - \vec{k}'} U_{\text{eff}}(\vec{k}'' - \vec{k}) \quad \text{Viewing } \vec{k} \text{ states as discrete. See Eq. (6.12)} \quad (27.108)$$

The fact that the interaction depends only upon the distance $\vec{r}_1 - \vec{r}_2$ between particles enforces conservation of momentum when viewed in $\vec{k}$ space. Thus, returning to Eq. (27.104), and including electron spin, one has the effective interaction

$$\hat{\mathcal{H}} = \frac{1}{2} \frac{1}{V} \sum_{\vec{q}\vec{k}\vec{k}'} \frac{4\pi e^2}{q^2 + \kappa_c^2} \left[1 + \frac{\omega_{\vec{q}}^2 - \bar{\omega}_{\vec{q}}^2}{\omega^2 - \omega_{\vec{q}}^2} \right] \hat{c}_{\vec{k}\sigma}^\dagger \hat{c}_{\vec{k}'\sigma'}^\dagger \hat{c}_{\vec{k} - \vec{q}\sigma} \hat{c}_{\vec{k}' + \vec{q}\sigma'}. \quad (27.109)$$

27.3.2 Instability of the Normal State: Cooper Problem

Cooper (1956) performed the first calculation showing how a small attractive interaction between electrons destabilizes the conventional electronic ground state. The idea is the following. Consider a state containing N particles that occupy $\vec{k}$ states up to a Fermi level. This state will be denoted by $|G\rangle$, and the first guess is that it is the ground state of the Hamiltonian, or at least close to it. Formally,

$$|G\rangle = \prod_{k < k_F} c_{\vec{k}}^\dagger |\emptyset\rangle. \quad (27.110)$$

Now imagine adding two new particles to $|G\rangle$. One possibility is that the two new particles seek out the lowest-lying unfilled $\vec{k}$ states and occupy them. Solutions of this sort certainly exist. But in the case where the particles interact by an attractive potential, there is another possibility; they may form a bound state that in no respect resembles the independent particles. Such a Cooper pair is at the heart of superconductivity. The calculation will show that states of the form (27.110) are unstable. While identification of an instability is never sufficient to show where the unstable system will terminate its travels, the nature of the instability also gives sufficient clues that the problem can be recast and solved completely.

Consider first a simplified treatment that leaves open a few nagging questions, but makes the development much clearer than the proper formalism. The Schrödinger equation for two electrons added on top of a Fermi sea of N electrons, occupying $\vec{k}$ states up to k_F is

$$\left[\frac{-\hbar^2 \nabla_1^2}{2m} + \frac{-\hbar^2 \nabla_2^2}{2m} + U(\vec{r}_1 - \vec{r}_2) \right] \Psi(\vec{r}_1, \vec{r}_2) = \mathcal{E} \Psi(\vec{r}_1, \vec{r}_2), \quad (27.111)$$

subject to the constraint that $\Psi(\vec{r}_1, \vec{r}_2)$ contain no Fourier components with $\vec{k}$ less than k_F . To simplify matters, assume that the pair of electrons described by Ψ has no net momentum and can therefore be written in the form

$$\Psi(\vec{r}_1, \vec{r}_2) = \sum_{k' > k_F} \Psi_{\vec{k}'} e^{-i\vec{k}' \cdot (\vec{r}_1 - \vec{r}_2)}. \quad (27.112)$$

Placing Eq. (27.112) into Eq. (27.111), multiplying by $\exp[i\vec{k} \cdot (\vec{r}_1 - \vec{r}_2)]$ and integrating over $\vec{r}_1 - \vec{r}_2$ gives

$$(2\epsilon_{\vec{k}} - \mathcal{E}) \Psi_{\vec{k}} + \sum_{k' > k_F} U_{\vec{k}\vec{k}'} \Psi_{\vec{k}'} = 0. \quad (27.113)$$

$\epsilon_{\vec{k}} = \hbar^2 k^2 / 2m$ is the energy of a single-particle state. The potential $U_{\vec{k}\vec{k}'}$ is $\langle \vec{k} | \hat{U} | \vec{k}' \rangle$. Because $|\vec{k}\rangle$ is a normalized plane wave $\exp[i\vec{k} \cdot \vec{r}] / \sqrt{\mathcal{V}}$, $U_{\vec{k}\vec{k}'}$ is $1/\mathcal{V}$ times the conventional Fourier transform of the potential $U(\vec{r})$.

In order to make further analytical progress, one must make some assumption about the form of U . An assumption that is particularly suited to analytical progress is

$$U_{\vec{k}\vec{k}'} = -\frac{U_0}{\mathcal{V}} \theta(\hbar\omega - |\mathcal{E}_F - \epsilon_{\vec{k}}|) \theta(\hbar\omega - |\mathcal{E}_F - \epsilon_{\vec{k}'}|). \quad (27.114)$$

Technically, the form of potential U assumed in Eq. (27.111) is not consistent with this sort of behavior in $\vec{k}$ space. One could retroactively modify Eq. (27.111) to avoid being upset by this. The constant U_0 has dimensions of energy times volume, and it is derived from the Fourier transform of the potential $U(\vec{r})$.

When one lets $k_{\max}$ be the largest value of k for which $\hbar\omega - |\mathcal{E}_F - \epsilon_{\vec{k}}|$ is positive, Eq. (27.113) takes the form

$$(2\epsilon_{\vec{k}} - \mathcal{E})\Psi_{\vec{k}} = \frac{U_0}{\mathcal{V}} \sum_{\vec{k}' > k_F}^{k_{\max}} \Psi_{\vec{k}'}, \quad \text{True so long as } k < k_{\max}; \text{ otherwise the right hand side vanishes.} \quad (27.115)$$

One can construct a host of solutions to Eq. (27.115) by picking any two wave vectors $\vec{k}_a$ and $\vec{k}_b$ whose magnitudes (above k_F) are the same, setting

$$\mathcal{E} = 2\epsilon_{\vec{k}_a}, \quad (27.116)$$

and taking the only nonzero components of Ψ to be

$$\Psi_{\vec{k}_a} = -\Psi_{\vec{k}_b} = \frac{1}{\sqrt{2}}. \quad (27.117)$$

Solutions of this type correspond to adding two particles to a normal Fermi liquid. If, however, one assumes that $\mathcal{E} - 2\epsilon_{\vec{k}} \neq 0$, then by summing Eq. (27.115) over $\vec{k}$ one gets

$$\sum_{\vec{k} > k_F}^{k_{\max}} \Psi_{\vec{k}} = \sum_{\vec{k} > k_F}^{k_{\max}} \frac{U_0}{\mathcal{V}} \frac{1}{(2\epsilon_{\vec{k}} - \mathcal{E})} \sum_{\vec{k}' > k_F}^{k_{\max}} \Psi_{\vec{k}'}. \quad (27.118)$$

$$\Rightarrow 1 = \sum_{\vec{k}}^{k_{\max}} \frac{U_0}{(2\epsilon_{\vec{k}} - \mathcal{E})\mathcal{V}} \quad (27.119)$$

$$\approx \int_{\mathcal{E}_F}^{\mathcal{E}_F + \hbar\omega} d\epsilon \frac{D(\mathcal{E}_F)}{2} \frac{U_0}{2\epsilon - \mathcal{E}} \quad \text{The factor of } 1/2 \text{ is needed because there is no sum over spin.} \quad (27.120)$$

$$\Rightarrow 1 = \frac{1}{4} D(\mathcal{E}_F) U_0 \ln\left(\frac{2\mathcal{E}_F + 2\hbar\omega - \mathcal{E}}{2\mathcal{E}_F - \mathcal{E}}\right). \quad (27.121)$$

Assuming that $U_0 D(\mathcal{E}_F)$ is much less than one, an approximate solution of Eq. (27.121) is

$$\mathcal{E} = 2\mathcal{E}_F - (2\mathcal{E}_{\max} - 2\mathcal{E}_F) \exp\left[-\frac{4}{D(\mathcal{E}_F)U_0}\right]. \quad (27.122)$$

The energy of the bound state is only slightly smaller than the energy of two additional particles at the Fermi level. However, the wave function is quite different from the form in Eq. (27.117); it is

$$\Psi_{\vec{k}} = \frac{U_0}{(2\epsilon_{\vec{k}} - \mathcal{E})\mathcal{V}} \sum_{\vec{k}' > k_F}^{k_{\max}} \Psi_{\vec{k}'}. \quad (27.123)$$

This wave function is rotationally invariant. Because the two electrons making up the wave function must be antisymmetric, they must be multiplied by an antisymmetric spin wave function in order to satisfy the Pauli principle. The conclusion

is that two electrons in an antisymmetric spin state, and with opposite $\vec{k}$ vectors so as to have no net momentum, can form a bound pair if they are added on top of collection of other electrons that are assumed not to have formed pairs of this type.

But then the assumption that the other electrons have not already formed bound pairs is clearly wrong, and one has to start all over at the beginning. However, before doing so, it is worth recasting the Cooper problem in the language of second quantization, to set the stage for the Hamiltonian actually used to study superconductivity.

27.3.3 Self-Consistent Ground State

Model Hamiltonian. Given the observation that pairs of electrons with opposite wave vectors and spins seem capable of binding into pairs, Bardeen, Cooper, and Schrieffer (1957) proposed the *BCS model Hamiltonian*, which abstracts from all the preceding complicated discussion a simple solvable model. It is

$$\hat{\mathcal{H}}_{\text{BCS}} = \sum_{\vec{k}, \sigma} \epsilon_{\vec{k}} \hat{c}_{\vec{k}\sigma}^\dagger \hat{c}_{\vec{k}\sigma} + \sum_{\vec{k}\vec{k}'} U_{\vec{k}\vec{k}'} \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{-\vec{k}\downarrow}^\dagger \hat{c}_{-\vec{k}'\downarrow} \hat{c}_{\vec{k}'\uparrow}. \quad (27.124)$$

A glaring departure from Eq. (27.109) is the absence of the sum on $\vec{q}$. Nonzero values of $\vec{q}$ correspond to electron pairs whose center of mass momentum is nonzero. These play no role in the ground state, so the $\vec{q}$ sum can be omitted from this Hamiltonian. Later, when interaction with external fields is considered, the $\vec{q}$ sum will have to be brought back.

The matrix elements $U_{\vec{k}\vec{k}'}$ will not be needed for some time. In model calculations they can freely be altered into any form that decays for large $\vec{k}$ and that is negative for some $\vec{k}$ and $\vec{k}'$ near the Fermi surface. With use of Eq. (27.114), the problem can be solved exactly, and when need arises, this form of the potential will be adopted.

The Cooper problem suggests that the Fermi sea is unstable toward the creation of correlated pairs of electrons. How can one create a state that has definite numbers of electrons correlated in pairs? A simple guess is that the state is of the form

$$|\Phi_N\rangle = \left[\sum_{\vec{k}} \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{-\vec{k}\downarrow}^\dagger g_{\vec{k}} \right]^N |\emptyset\rangle, \quad (27.125)$$

which creates all possible pairs of $2N$ particles with various weights. If $g_{\vec{k}} = 1$ up to the Fermi surface and is zero thereafter, then the state Φ_N is precisely a Bloch state. The only terms in the product to survive are those that create precisely one sample of each type of particle; there are $N!$ such terms, so one must divide through by this factor to normalize the state. Bardeen, Cooper, and Schrieffer found that for formal reasons this type of state is inconvenient to work with. The difficulties are similar to those that arise in classical statistical mechanics. They make it advantageous to work in the grand canonical rather than the canonical ensemble, and are ameliorated by considering

$$|\Phi\rangle \equiv \sum_N \frac{1}{N!} |\Phi_N\rangle \quad (27.126)$$

$$= \sum_N \frac{1}{N!} \left[\sum_{\vec{k}} \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{-\vec{k}\downarrow}^\dagger g_{\vec{k}} \right]^N |\Phi\rangle. \quad (27.127)$$

The constants $g_{\vec{k}}$ could also depend upon N . Neglecting this dependence is justified by noting that the sum will be dominated by a narrow range of N and assuming that $g_{\vec{k}}$ does not vary appreciably with N in this range.

Coherent States. Equation (27.127) can be rewritten as

$$|\Phi\rangle = \exp\left[\sum_{\vec{k}} g_{\vec{k}} \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{-\vec{k}\downarrow}^\dagger\right] |\Phi\rangle. \quad (27.128)$$

(27.128) is a *coherent state*, similar to the states that allow one to find the classical limit of quantum optics. Coherent states have pleasant formal properties that offset the fact they have indefinite particle number. For fermion operators, only the zero and first powers survive in the exponential, so one can finally rewrite Eq. (27.128) as

$$|\Phi\rangle = \prod_{\vec{k}} \left[1 + g_{\vec{k}} \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{-\vec{k}\downarrow}^\dagger \right] |\Phi\rangle \equiv \hat{\Phi} |\Phi\rangle. \quad (27.129)$$

The variational procedure of Section B.2 dictates that given a trial state and a Hamiltonian, one has to take the expectation value of the Hamiltonian and try to minimize it with respect to all of the parameters in the trial wave function. The wave function does not have definite particle number, which means that instead of enforcing the constraint that $|\Phi\rangle$ be normalized, it is necessary to enforce the constraint that the average particle number be maintained at N . This constraint is maintained by introducing a Lagrange multiplier that is easily identified as the chemical potential μ .

Here are several identities that make it possible to manipulate the coherent state.

1. The normalization of the state is given by

$$\langle \Phi | \Phi \rangle = \prod_{\vec{k}} (1 + |g_{\vec{k}}|^2) = \mathcal{N}^2. \quad (27.130)$$

2. Taking the expectation of the operators which destroy a pair of electrons, one obtains

$$b_{\vec{k}} = \frac{1}{\mathcal{N}^2} \langle \Phi | \hat{c}_{-\vec{k}\downarrow} \hat{c}_{\vec{k}\uparrow} | \Phi \rangle = \frac{g_{\vec{k}}}{1 + |g_{\vec{k}}|^2}, \quad (27.131)$$

because one of the factors of $1 + |g_{\vec{k}}|^2$ disappears and is replaced by $g_{\vec{k}}$. Similarly,

$$\frac{1}{\mathcal{N}^2} \langle \Phi | \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{-\vec{k}\downarrow}^\dagger \hat{c}_{-\vec{k}'\downarrow} \hat{c}_{\vec{k}'\uparrow} | \Phi \rangle = b_{\vec{k}}^* b_{\vec{k}'}. \quad (27.132)$$

3. The number of particles of a given spin in the wave function equals the number of pairs containing that particle: formally, one has

$$\left[\sum_{\sigma} \hat{n}_{\vec{k}\sigma}, \hat{\Phi} \right] = \left[g_{\vec{k}} \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{-\vec{k}\downarrow}^\dagger + g_{-\vec{k}} \hat{c}_{-\vec{k}\uparrow}^\dagger \hat{c}_{\vec{k}\downarrow}^\dagger \right] \hat{\Phi}. \quad (27.133)$$

Expectation Value of Hamiltonian. Identities (27.130) through (27.133) make it possible to evaluate the expectation value of the Hamiltonian (27.124). The kinetic energy expectation value comes from Eq. (27.133), and then Eq. (27.131):

$$\frac{1}{\mathcal{N}^2} \langle \Phi | \sum_{\sigma} \hat{n}_{\vec{k}\sigma} | \Phi \rangle = \frac{1}{\mathcal{N}^2} \langle \Phi | \left(g_{\vec{k}} \hat{c}_{\vec{k}\uparrow}^{\dagger} \hat{c}_{-\vec{k}\downarrow}^{\dagger} + g_{-\vec{k}} \hat{c}_{-\vec{k}\uparrow}^{\dagger} \hat{c}_{\vec{k}\downarrow}^{\dagger} \right) | \Phi \rangle \quad (27.134)$$

$$\Rightarrow \sum_{\sigma} n_{\vec{k}\sigma} = g_{\vec{k}} b_{\vec{k}}^* + g_{-\vec{k}} b_{-\vec{k}}^*. \quad (27.135)$$

Equation (27.132) already records the matrix element needed for the potential term in the BCS Hamiltonian. Therefore, the expectation value of Eq. (27.124), including the Lagrange multiplier μ to enforce particle number, is

$$\langle \Phi | \hat{\mathcal{H}}_{\text{BCS}} - \mu N | \Phi \rangle = \sum_{\vec{k}} 2 \left(\epsilon_{\vec{k}} - \mu \right) g_{\vec{k}} b_{\vec{k}}^* + \sum_{\vec{k}\vec{k}'} U_{\vec{k}\vec{k}'} b_{\vec{k}}^* b_{\vec{k}'}. \quad (27.136)$$

Varying Parameters in Trial State. In order to minimize Eq. (27.136) with respect to the parameters $g_{\vec{k}}$, take the derivative with respect to $g_{\vec{q}}^*$. Noting that

$$\frac{\partial b_{\vec{k}}^*}{\partial g_{\vec{k}}^*} = \frac{1}{(1 + |g_{\vec{k}}|^2)^2}; \quad \frac{\partial b_{\vec{k}}}{\partial g_{\vec{k}}^*} = -\frac{g_{\vec{k}}^2}{(1 + |g_{\vec{k}}|^2)^2}, \quad (27.137)$$

one has

$$\frac{2(\epsilon_{\vec{q}} - \mu)g_{\vec{q}}}{(1 + |g_{\vec{q}}|^2)^2} + \sum_{\vec{k}\vec{k}'} \frac{U_{\vec{k}\vec{k}'}}{(1 + |g_{\vec{q}}|^2)^2} \left[b_{\vec{k}'} \delta_{\vec{k}\vec{q}} - b_{\vec{k}}^* g_{\vec{q}}^2 \delta_{\vec{q}\vec{k}'} \right] = 0. \quad (27.138)$$

It is conventional to define the *gap function*

$$\Delta_{\vec{k}} = - \sum_{\vec{k}'} U_{\vec{k}\vec{k}'} b_{\vec{k}'}, \quad (27.139)$$

in terms of which Eq. (27.138) becomes

$$0 = 2 (\epsilon_{\vec{q}} - \mu) g_{\vec{q}} - \Delta_{\vec{q}} + g_{\vec{q}}^2 \Delta_{\vec{q}}^* \quad \text{Use the fact that } U_{\vec{k}\vec{k}'} = U_{\vec{k}'\vec{k}}^*. \quad (27.140)$$

$$\Rightarrow g_{\vec{k}} = \frac{\mathcal{E}_{\vec{k}} - (\epsilon_{\vec{k}} - \mu)}{\Delta_{\vec{k}}^*}, \quad (27.141)$$

with

$$\mathcal{E}_{\vec{k}} = \sqrt{(\epsilon_{\vec{k}} - \mu)^2 + |\Delta_{\vec{k}}|^2}. \quad (27.142)$$

The energy $\mathcal{E}_{\vec{k}}$ will emerge as the energy of an excitation above the ground state indexed by $\vec{k}$. It is always positive, and is shown in Figure 27.10. With a little algebra, it follows that

$$b_{\vec{k}} = \frac{g_{\vec{k}}}{1 + |g_{\vec{k}}|^2} = \frac{\Delta_{\vec{k}}}{2\mathcal{E}_{\vec{k}}}. \quad (27.143)$$

Chemical Potential. To calculate physical quantities, it is necessary to determine the chemical potential μ . Despite the fact that electrons occupy a complicated correlated state, the Fermi energy can still be defined by

$$N = 2 \sum_{\vec{k}} \theta(\mathcal{E}_F - \epsilon_{\vec{k}}) = \int_0^{\mathcal{E}_F} d\epsilon D(\epsilon), \quad (27.144)$$

and the chemical potential turns out to equal $\mathcal{E}_F$. To see why, recall that the chemical potential is determined implicitly from the constraint

$$N = \sum_{\vec{k}\sigma} g_{\vec{k}}^* b_{\vec{k}} = \sum_{\vec{k}\sigma} \frac{1}{2} \left[1 - \frac{\epsilon_{\vec{k}} - \mu}{\mathcal{E}_{\vec{k}}} \right] \quad \text{See Eq. (27.135), and Figure 27.10.} \quad (27.145)$$

$$= \sum_{\vec{k}\sigma} \frac{1}{2} \left[1 - \frac{\epsilon_{\vec{k}} - \mu}{\sqrt{(\epsilon_{\vec{k}} - \mu)^2 + |\Delta|^2}} \right] \quad (27.146)$$

$$= \int d\epsilon \frac{D(\epsilon)}{2} \left[1 - \frac{\epsilon - \mu}{\sqrt{(\epsilon - \mu)^2 + |\Delta|^2}} \right] \quad (27.147)$$

$$= \int d\epsilon \left[\int^\epsilon d\epsilon' D(\epsilon') \right] \frac{1}{2} \frac{\partial}{\partial \epsilon} \frac{\epsilon - \mu}{\sqrt{(\epsilon - \mu)^2 + |\Delta|^2}} \quad (27.148)$$

$$= \int d\epsilon \left[\int^\mu d\epsilon' D(\epsilon') \right] \frac{1}{2} \frac{\partial}{\partial \epsilon} \frac{\epsilon - \mu}{\sqrt{(\epsilon - \mu)^2 + |\Delta|^2}} + \mathcal{O}(\Delta/\mathcal{E}_F)^2 \quad (27.149)$$

See the Sommerfeld expansion in Section 6.5.

$$= \left[\int^\mu d\epsilon' D(\epsilon') \right] = N + D(\mathcal{E}_F)(\mu - \mathcal{E}_F) \quad (27.150)$$

$$\Rightarrow \mu = \mathcal{E}_F. \quad \text{Use Eq. (27.144).} \quad (27.151)$$

Gap Equation. From the definition of $\Delta_{\vec{k}}$ it follows that

$$\Delta_{\vec{k}} = - \sum_{\vec{k}'} U_{\vec{k}\vec{k}'} \frac{\Delta_{\vec{k}'}}{2\mathcal{E}_{\vec{k}'}}. \quad (27.152)$$

Equation (27.152) is the *gap equation* at zero temperature. It can be solved analytically if one assumes the simple separable potential given in Eq. (27.114). Then

$$\Delta_{\vec{k}} = \theta(\hbar\omega - |\epsilon_{\vec{k}} - \mathcal{E}_F|) \frac{U_0}{V} \sum_{\vec{k}'} \frac{\Delta_{\vec{k}'}}{2\sqrt{(\epsilon_{\vec{k}'} - \mu)^2 + |\Delta_{\vec{k}'}|^2}}. \quad (27.153)$$

The step function inside the sum need not be written out explicitly, because the functional form of $\Delta_{\vec{k}}$ is

$$\Delta_{\vec{k}} = \Delta \theta(\hbar\omega - |\epsilon_{\vec{k}} - \mathcal{E}_F|). \quad (27.154)$$

To determine the remaining constant Δ , assume that $\Delta_{\vec{k}}$ is not zero for all $\vec{k}$, so that

$$1 = \sum_{\vec{k}} \frac{1}{V} \theta(\hbar\omega - |\epsilon_{\vec{k}} - \mathcal{E}_F|) \frac{U_0}{2\sqrt{(\epsilon_{\vec{k}} - \mu)^2 + |\Delta|^2}}. \quad (27.155)$$

$$= \int_{\mathcal{E}_F - \hbar\omega}^{\mathcal{E}_F + \hbar\omega} d\epsilon \frac{D(\epsilon)}{2} \frac{U_0}{2\sqrt{(\epsilon - \mathcal{E}_F)^2 + |\Delta|^2}} \quad \begin{array}{l} \text{Because } \mu = \mathcal{E}_F. \text{ Factor of } 1/2 \\ \text{in density of states because there} \\ \text{is no spin sum.} \end{array} \quad (27.156)$$

$$= U_0 \int_0^{\hbar\omega/\Delta} d\zeta \frac{D(\mathcal{E}_F)}{2\sqrt{\zeta^2 + 1}} \quad (27.157)$$

$$= \frac{U_0 D(\mathcal{E}_F)}{2} \sinh^{-1} \frac{\hbar\omega}{\Delta} \quad (27.158)$$

$$\Rightarrow \Delta = 2\hbar\omega \exp \left[-\frac{2}{D(\mathcal{E}_F)U_0} \right]. \quad \begin{array}{l} \text{Correct when } D(\mathcal{E}_F)U_0 \ll 1. \text{ Frequently, } D(\mathcal{E}_F) \\ \text{is denoted instead by } 2N(0), \text{ where } N(0) \text{ is} \\ \text{the density of states of a single spin direction} \\ \text{at the Fermi surface.} \end{array} \quad (27.159)$$

The binding energy Δ that emerges from this calculation differs from the one that appeared in Eq. (27.122) by a factor of 2 inside the exponential. This binding energy is exponentially larger than the binding of an isolated Cooper pair, because all N particles in the wave function are participating in the bound state in a consistent way.

27.3.4 Thermodynamics of Superconductors

The great advantage of working in the grand canonical ensemble is most evident when one wants to do thermodynamics. The grand partition function is

$$Z_{\text{gr}} = \text{Tre}^{-\beta[\hat{\mathcal{H}}_{\text{BCS}} - \mu\hat{N}]}, \quad (27.160)$$

where the trace is over all sets of Fermi occupation numbers. Because $\hat{\mathcal{H}}_{\text{BCS}}$ is quartic in Fermi operators, the trace cannot be performed in closed form. However, one can get an excellent approximation to the thermodynamics employing a version of mean field theory that is designed to reproduce the results already obtained from coherent states at zero temperature. The expectation value $b_{\vec{k}} = \langle \hat{c}_{-\vec{k}\downarrow} \hat{c}_{\vec{k}\uparrow} \rangle$ is all one needs to understand the ground state. So guess that mean field theory is accomplished by setting

$$\hat{c}_{-\vec{k}\downarrow} \hat{c}_{\vec{k}\uparrow} = b_{\vec{k}} + \left(\hat{c}_{-\vec{k}\downarrow} \hat{c}_{\vec{k}\uparrow} - b_{\vec{k}} \right), \quad (27.161)$$

treating the second term as small, and expanding to first order only in the small quantities. The result of carrying out this procedure is the effective grand partition function

$$Z_{\text{gr}} = \text{Tre}^{-\beta[\hat{\mathcal{H}}_{\text{eff}} - \mu N]}, \quad (27.162)$$

where

$$\begin{aligned} \hat{\mathcal{H}}_{\text{eff}} - \mu N &= \sum_{\vec{k}\sigma} \hat{n}_{\vec{k}\sigma} (\epsilon_{\vec{k}} - \mu) + \sum_{\vec{k}\vec{k}'} b_{\vec{k}'} U_{\vec{k}\vec{k}'} \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{-\vec{k}\downarrow}^\dagger + b_{\vec{k}}^* U_{\vec{k}'\vec{k}}^* \hat{c}_{-\vec{k}'\downarrow} \hat{c}_{\vec{k}\uparrow} - b_{\vec{k}}^* b_{\vec{k}'} U_{\vec{k}\vec{k}'} \quad (27.163) \\ &\equiv \sum_{\vec{k}\sigma} \hat{n}_{\vec{k}\sigma} (\epsilon_{\vec{k}} - \mu) - \sum_{\vec{k}} [\Delta_{\vec{k}} \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{-\vec{k}\downarrow}^\dagger + \Delta_{\vec{k}}^* \hat{c}_{-\vec{k}\downarrow} \hat{c}_{\vec{k}\uparrow}] - \sum_{\vec{k}\vec{k}'} b_{\vec{k}}^* b_{\vec{k}'} U_{\vec{k}\vec{k}'} \quad (27.164) \end{aligned}$$

In the ground state, this decomposition was exact; at nonzero temperature it is no longer exact, but provides a very good approximation except exceedingly close to the critical temperature. Because products of no more than two Fermi operators appear now in Eq. (27.164), the trace can be performed exactly by the means used to solve the ideal Fermi gas. First, however, it is necessary to diagonalize the Hamiltonian. The appropriate transformation was introduced by Bogoliubov (1958) and by Valatin (1958). Perform the canonical transformation

$$\hat{c}_{\vec{k}\uparrow} = u_{\vec{k}} \hat{\gamma}_{\vec{k}\uparrow} + v_{\vec{k}}^* \hat{\gamma}_{\vec{k}\downarrow}^\dagger \quad (27.165a)$$

$$\hat{c}_{-\vec{k}\downarrow}^\dagger = -v_{\vec{k}} \hat{\gamma}_{\vec{k}\uparrow} + u_{\vec{k}}^* \hat{\gamma}_{\vec{k}\downarrow}^\dagger. \quad (27.165b)$$

The new operators only obey the correct Fermi commutation relations if

$$|u_{\vec{k}}|^2 + |v_{\vec{k}}|^2 = 1. \quad (27.166)$$

Placing Eqs. (27.165) into Eq. (27.164), one finds

$$\hat{\mathcal{H}}_{\text{eff}} - \mu N = \sum_{\vec{k}} \left[(\epsilon_{\vec{k}} - \mu) \left\{ \begin{aligned} &[u_{\vec{k}}^* \hat{\gamma}_{\vec{k}\uparrow}^\dagger + v_{\vec{k}} \hat{\gamma}_{\vec{k}\downarrow}] [-u_{\vec{k}} \hat{\gamma}_{\vec{k}\uparrow} + v_{\vec{k}}^* \hat{\gamma}_{\vec{k}\downarrow}^\dagger] \\ &+ [-v_{\vec{k}} \hat{\gamma}_{\vec{k}\uparrow} + u_{\vec{k}}^* \hat{\gamma}_{\vec{k}\downarrow}^\dagger] [-v_{\vec{k}}^* \hat{\gamma}_{\vec{k}\uparrow}^\dagger + u_{\vec{k}} \hat{\gamma}_{\vec{k}\downarrow}] \end{aligned} \right\} \right. \\ \left. - \Delta_{\vec{k}} [u_{\vec{k}}^* \hat{\gamma}_{\vec{k}\uparrow}^\dagger + v_{\vec{k}} \hat{\gamma}_{\vec{k}\downarrow}] [-v_{\vec{k}} \hat{\gamma}_{\vec{k}\uparrow} + u_{\vec{k}}^* \hat{\gamma}_{\vec{k}\downarrow}^\dagger] \right. \\ \left. - \Delta_{\vec{k}}^* [-v_{\vec{k}}^* \hat{\gamma}_{\vec{k}\uparrow}^\dagger + u_{\vec{k}} \hat{\gamma}_{\vec{k}\downarrow}] [u_{\vec{k}} \hat{\gamma}_{\vec{k}\uparrow} + v_{\vec{k}}^* \hat{\gamma}_{\vec{k}\downarrow}^\dagger] \right. \\ \left. - \sum_{\vec{k}'} b_{\vec{k}'}^* b_{\vec{k}} U_{\vec{k}\vec{k}'} \right] \quad (27.167)$$

Requiring the coefficients of the nondiagonal terms, for example, of $\hat{\gamma}_{\vec{k}\downarrow} \hat{\gamma}_{\vec{k}\uparrow}^\dagger$, to vanish leads to the condition

$$2u_{\vec{k}} v_{\vec{k}} (\epsilon_{\vec{k}} - \mu) + \Delta_{\vec{k}} v_{\vec{k}}^2 - \Delta_{\vec{k}}^* u_{\vec{k}}^2 = 0. \quad (27.168)$$

One has the freedom to choose one of $u_{\vec{k}}$ or $v_{\vec{k}}$ real; choose $u_{\vec{k}}$, so as to write

$$0 = 2\sqrt{1 - |v_{\vec{k}}|^2} v_{\vec{k}} (\epsilon_{\vec{k}} - \mu) + \Delta_{\vec{k}} v_{\vec{k}}^2 - \Delta_{\vec{k}}^* (1 - |v_{\vec{k}}|^2). \quad (27.169)$$

Write $v_{\vec{k}}$ as

$$v_{\vec{k}} = \frac{g_{\vec{k}}^*}{\sqrt{1 + |g_{\vec{k}}|^2}}, \quad (27.170)$$

thus defining $g_{\vec{k}}$. Then Eq. (27.169) becomes

$$0 = 2(\epsilon_{\vec{k}} - \mu) g_{\vec{k}} - \Delta_{\vec{k}} + g_{\vec{k}}^2 \Delta_{\vec{k}}^*. \quad (27.171)$$

So the definition of $g_{\vec{k}}$ is no accident: It satisfies Eq. (27.140) precisely as before. Once this condition is met, the Hamiltonian becomes diagonal. As before,

$$g_{\vec{k}} = \frac{\mathcal{E}_{\vec{k}} - (\epsilon_{\vec{k}} - \mu)}{\Delta_{\vec{k}}^*}. \quad (27.172)$$

For future reference, note that

$$|v_{\vec{k}}|^2 = \frac{\mathcal{E}_{\vec{k}} - (\epsilon_{\vec{k}} - \mu)}{2\mathcal{E}_{\vec{k}}}, \quad |u_{\vec{k}}|^2 = \frac{\mathcal{E}_{\vec{k}} + \epsilon_{\vec{k}} - \mu}{2\mathcal{E}_{\vec{k}}}, \quad v_{\vec{k}}u_{\vec{k}}^* = \frac{\Delta_{\vec{k}}^*}{2\mathcal{E}_{\vec{k}}}. \quad (27.173)$$

Writing out the Hamiltonian in terms of the new operators $\hat{\gamma}$, one finds that

$$\hat{\mathcal{H}}_{\text{eff}} - \mu N = \sum_{\vec{k}} \mathcal{E}_{\vec{k}} \left[\hat{\gamma}_{\vec{k}\uparrow}^\dagger \hat{\gamma}_{\vec{k}\uparrow} + \hat{\gamma}_{\vec{k}\downarrow}^\dagger \hat{\gamma}_{\vec{k}\downarrow} \right] + \sum_{\vec{k}} \left[\epsilon_{\vec{k}} - \mu - \mathcal{E}_{\vec{k}} - \sum_{\vec{k}'} b_{\vec{k}}^* b_{\vec{k}'} U_{\vec{k}\vec{k}'} \right]. \quad (27.174)$$

This expression clearly shows that $\mathcal{E}_{\vec{k}}$ give the energies of excitations above the ground state.

There is still one piece of the puzzle missing, because $\Delta_{\vec{k}}$ has been defined in terms of $b_{\vec{k}}$, but nothing has yet been done to fix $b_{\vec{k}}$. Write

$$b_{-\vec{k}}^* = b_{\vec{k}}^* = \langle \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{-\vec{k}\downarrow}^\dagger \rangle = \langle v_{\vec{k}} u_{\vec{k}}^* (\hat{\gamma}_{\vec{k}\downarrow}^\dagger \hat{\gamma}_{\vec{k}\downarrow}^\dagger - \hat{\gamma}_{\vec{k}\uparrow}^\dagger \hat{\gamma}_{\vec{k}\uparrow}^\dagger) \rangle + \text{terms with } \gamma\gamma \text{ or } \gamma^\dagger \gamma^\dagger. \quad (27.175)$$

This expectation value must be identical with what one expects for noninteracting fermions, because that is what Eq. (27.174) describes. Therefore

$$b_{\vec{k}}^* = v_{\vec{k}} u_{\vec{k}}^* (1 - 2f_{\vec{k}}), \quad (27.176)$$

where

$$f_{\vec{k}} = \frac{1}{e^{\beta \mathcal{E}_{\vec{k}}} + 1} \quad (27.177)$$

is the Fermi function. However, it is an unusual Fermi function. The denominator features $\exp[\beta \mathcal{E}_{\vec{k}}]$, rather than $\exp[\beta(\mathcal{E}_{\vec{k}} - \mu)]$. The formal reason is that although μ is buried inside the definition of $\mathcal{E}_{\vec{k}}$, it does not multiply the quantum operators in Eq. (27.174). Physically, the reason is that $\hat{\gamma}^\dagger$ creates an excited state, but does not change particle number. Because $\mathcal{E}_{\vec{k}}$ is always positive, $f_{\vec{k}}$ reaches its maximum value at the Fermi surface, and it decays to zero when $\vec{k}$ lies either above or below, as shown in Figure 27.10. It describes the occupation probability of holes below the Fermi surface and electrons above the Fermi surface.

Using Eqs. (27.170), (27.166), and (27.172) to determine $u_{\vec{k}}$ and $v_{\vec{k}}$ gives finally that

$$b_{\vec{k}} = \frac{\Delta_{\vec{k}}}{2\mathcal{E}_{\vec{k}}} (1 - 2f_{\vec{k}}) \quad (27.178)$$

$$\Rightarrow \sum_{\vec{k}'} b_{\vec{k}'} U_{\vec{k}\vec{k}'} = -\Delta_{\vec{k}} = \sum_{\vec{k}'} \frac{\Delta_{\vec{k}'}}{2\mathcal{E}_{\vec{k}'}} (1 - 2f_{\vec{k}'}) U_{\vec{k}\vec{k}'}, \quad (27.179)$$

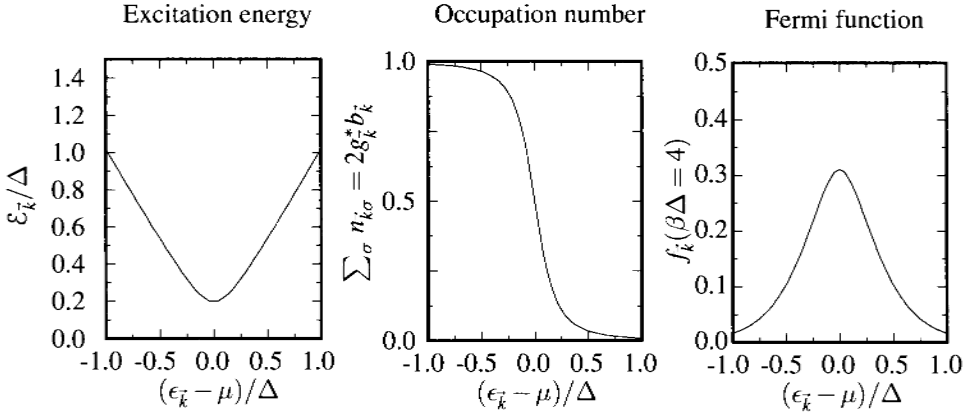


Figure 27.10. Sketches of the excitation energy, occupation number, and Fermi function for the BCS theory of superconductivity.

generalizing the BCS gap equation to nonzero temperatures.

The critical temperature is the highest one for which this equation has a solution for Δ nonzero. So the critical temperature should be given by the solution of the gap equation when Δ is infinitesimal. Using the same simple form of $U_{\vec{k}\vec{k}'}$ from Eq. (27.114) shows that if Δ is almost zero then

$$\Delta = \sum_{\vec{k}} \theta(\hbar\omega - |\mathcal{E}_F - \epsilon_{\vec{k}}|) \frac{\Delta}{2|\epsilon_{\vec{k}} - \mu|} \frac{U_0}{V} (1 - 2f_{\vec{k}}) \quad (27.180)$$

$$\Rightarrow 1 = \int_0^{\beta\hbar\omega} U_0 \frac{D(\mathcal{E}_F)}{2} \frac{dx}{x} \left[1 - \frac{2}{e^x + 1} \right] \quad x = \beta(\epsilon_{\vec{k}} - \mu). \quad (27.181)$$

$$\approx \frac{U_0 D(\mathcal{E}_F)}{2} \left\{ \ln \beta\hbar\omega \left[1 - \frac{2}{e^{\beta\hbar\omega} + 1} \right] + 2 \int_0^{\infty} dx \ln x \frac{\partial}{\partial x} \frac{1}{e^x + 1} \right\} \quad (27.182)$$

$$\approx \frac{U_0 D(\mathcal{E}_F)}{2} \left\{ \ln(\beta\hbar\omega) + \ln\left(\frac{2\gamma_E}{\pi}\right) \right\}, \quad \begin{array}{l} \text{Computations use} \\ \text{approximation } \beta\hbar\omega \gg 1; \\ \text{Eq. (27.184) shows that this} \\ \text{is the same as} \\ U_0 D(\mathcal{E}_F) \ll 1. \end{array} \quad (27.183)$$

where γ_E is Euler's constant. The final result is that

$$k_B T_c = \hbar\omega \frac{2\gamma_E}{\pi} \exp\left[-\frac{2}{U_0 D(\mathcal{E}_F)}\right], \quad (27.184)$$

and comparing with Eq. (27.159) predicts that the ratio between the gap at zero temperature and the critical temperature takes the universal value

$$\Rightarrow \frac{2\Delta(T=0)}{k_B T_c} = \frac{2\pi}{\gamma_E} = 3.53. \quad (27.185)$$

When the gap Δ is measured by tunneling, this relationship is found to hold to about 30 %, displayed in Table 27.2. Considering the simplicity of the form assumed for $U_{\vec{k}\vec{k}'}$, the agreement is quite acceptable.

Table 27.2. Thermodynamic properties of a variety of superconducting materials

Element	$2\Delta/k_B T$	$(C_s - C_n)/C_n$	Element	$2\Delta/k_B T$	$(C_s - C_n)/C_n$
BCS	3.53	1.43			
Al	2.5–4.2	1.3–1.6	Pb	4.0–4.4	2.7
Cd	3.2–3.4	1.3–1.4	Sn	2.8–4.0	1.6
Ga	3.5	1.4	Ta	3.5–3.7	1.6
Hg	4.0–4.6	2.4	Tl	3.6–3.9	1.5
In	3.4–3.7	1.7	V	3.4–3.5	1.5
La	1.7–3.2	1.5	Zn	3.2–3.4	1.2–1.3
Nb	3.6–3.8	1.9–2.0			

Ratio of energy gap Δ to critical temperature $k_B T$ [Eq. (27.185)] as well as fractional change in specific heat from normal to superconducting state is compared with predictions of Bardeen, Cooper, and Schrieffer. Source: Meservey and Schwartz (1969).

27.3.5 Superconductor in External Magnetic Field

Bogoliubov Hamiltonian. In order to find the behavior of a superconductor placed in an external magnetic field, the model Hamiltonian must be altered: made more complex in one respect, and simpler in another. The complication arises in the form of the electron interactions. Comparing Eqs. (27.109) and (27.124), one sees that a sum over $\vec{q}$ was dropped in moving from the first to the second. The interpretation of $\vec{q}$ is that it corresponds to the center of mass momentum of Cooper pairs. In the superconducting ground state, all the Cooper pairs are stationary, but this assumption is invalid when external fields are applied and supercurrents result. The sum over $\vec{q}$ must be added back in to the Hamiltonian. To keep formal complexity to the minimum, the potential $U_{\vec{k}\vec{k}'}$ will be simplified to the maximum degree and replaced by the constant value $-U_0/V$. Might such a choice lead to divergences? Equation (27.155) shows that it will not. Instead of that integral being cut off at $\mathcal{E}_F + \hbar\omega$, it would be cut off at $W/2$, where W is the bandwidth, and describes the range of energies where $D(\epsilon)$ is nonzero.

The formulation of superconductivity due to Bogoliubov (1958) therefore begins with the Hamiltonian

$$\hat{\mathcal{H}} = \sum_{\vec{k}\vec{k}'\sigma} \epsilon_{\vec{k}\vec{k}'} \hat{c}_{\vec{k}\sigma}^\dagger \hat{c}_{\vec{k}'\sigma} - \sum_{\vec{k}\vec{q}\vec{k}'} \frac{U_0}{V} \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{\vec{q}-\vec{k}\downarrow}^\dagger \hat{c}_{\vec{q}-\vec{k}'\downarrow} \hat{c}_{\vec{k}'\uparrow}. \quad (27.186)$$

The energy $\epsilon_{\vec{k}\vec{k}'}$ depends upon $\vec{k}$ and $\vec{k}'$ in order to accommodate the possible addition of external potentials.

The pattern by which quartic Hamiltonians are changed to quadratic ones should now be clear. One replaces the product of four fermion operators in the last term with products of two operators times the expectation values of the other two. Let

N_k denote the total number of plane waves $\vec{k}$, and define

$$\Delta_{\vec{q}} = \sum_{\vec{k}'} \frac{U_0}{V} \langle \hat{c}_{\vec{q}-\vec{k}'} \hat{c}_{\vec{k}'\uparrow} \rangle. \quad (27.187)$$

Then

$$\hat{\mathcal{H}} - \mu N = \sum_{\vec{k}\vec{k}'\sigma} [\epsilon_{\vec{k}\vec{k}'} - \mu \delta_{\vec{k}\vec{k}'}] \hat{c}_{\vec{k}\sigma}^\dagger \hat{c}_{\vec{k}'\sigma} - \sum_{\vec{k}\vec{q}} [\Delta_{\vec{q}}^* \hat{c}_{\vec{q}-\vec{k}\downarrow} \hat{c}_{\vec{k}\uparrow} + \Delta_{\vec{q}} \hat{c}_{\vec{k}\uparrow}^\dagger \hat{c}_{\vec{q}-\vec{k}\downarrow}^\dagger]. \quad (27.188)$$

There should be an extra constant subtracted off, as in the nonzero temperature calculation, but it can be absorbed into the one-particle energy.

In order to focus upon external potentials, the Hamiltonian must be written in terms of $\vec{r}$ rather than $\vec{k}$. Define

$$\hat{c}_{i\sigma} = \sum_{\vec{k}} \frac{e^{-i\vec{k}\cdot\vec{r}}}{\sqrt{N_k}} \hat{c}_{\vec{k}\sigma}, \quad \Delta_{\vec{k}} = \frac{1}{N_k} \sum_{\vec{r}} e^{i\vec{k}\cdot\vec{r}} \Delta_{\vec{r}}, \quad \epsilon_{\vec{k}\vec{k}'} = \frac{1}{N_k} \sum_{\vec{r}\vec{r}'} e^{i\vec{k}\cdot\vec{r} - i\vec{k}'\cdot\vec{r}'} \epsilon_{\vec{r}\vec{r}'}. \quad (27.189)$$

The factor of $1/\sqrt{N_k}$ guarantees that $\hat{c}_{\vec{r}}$ has the right anticommutation relations.

Substituting the final two relations in Eq. (27.189) into Eq. (27.188) and then using the first gives

$$\hat{\mathcal{H}} = \sum_{\vec{r}\vec{r}'\sigma} [\epsilon_{\vec{r}\vec{r}'} - \mu \delta_{\vec{r}\vec{r}'}] \hat{c}_{\vec{r}\sigma}^\dagger \hat{c}_{\vec{r}'\sigma} - \sum_{\vec{r}} [\Delta_{\vec{r}}^* \hat{c}_{\vec{r}\downarrow} \hat{c}_{\vec{r}\uparrow} + \Delta_{\vec{r}} \hat{c}_{\vec{r}\uparrow}^\dagger \hat{c}_{\vec{r}\downarrow}^\dagger]. \quad (27.190)$$

Diagonalizing the Hamiltonian. This Hamiltonian is quadratic, so in principle it can be diagonalized. Let $\gamma_{l\sigma}$ be a linear combination of the $\hat{c}$'s which diagonalizes the Hamiltonian. In other words, the γ operators are defined by

$$\hat{\mathcal{H}} = \sum_l \mathcal{E}_l [\hat{\gamma}_{l\uparrow}^\dagger \hat{\gamma}_{l\uparrow} + \hat{\gamma}_{l\downarrow}^\dagger \hat{\gamma}_{l\downarrow}]. \quad (27.191)$$

A sufficiently general transformation of the $\hat{c}$'s is of the form

$$\begin{aligned} \hat{c}_{\vec{r}\uparrow} &= \frac{1}{\sqrt{N_k}} \sum_l u_l(\vec{r}) \hat{\gamma}_{l\uparrow} + v_l^*(\vec{r}) \hat{\gamma}_{l\downarrow}^\dagger \\ \hat{c}_{\vec{r}\downarrow} &= \frac{1}{\sqrt{N_k}} \sum_l u_l(\vec{r}) \hat{\gamma}_{l\downarrow} - v_l^*(\vec{r}) \hat{\gamma}_{l\uparrow}^\dagger. \end{aligned} \quad (27.192)$$

The general transformation would involve four separate functions, but one finds in the course of the calculations that only two of them are independent, and that signs must be chosen in this way for the diagonalization to work.

In order to find equations for u and v note that

$$\begin{aligned} [\mathcal{H}_B, \hat{\gamma}_{l\sigma}] &= -\mathcal{E}_l \hat{\gamma}_{l\sigma} \\ [\mathcal{H}_B, \hat{\gamma}_{l\sigma}^\dagger] &= \mathcal{E}_l \hat{\gamma}_{l\sigma}^\dagger. \end{aligned} \quad \begin{array}{l} \text{These are commutators, not anticommutators,} \\ \text{and they work as indicated because the prod-} \\ \text{uct of two identical } \hat{\gamma}\text{'s is zero.} \end{array} \quad (27.193)$$

Next note that

$$[\mathcal{H}_B, \hat{c}_{\vec{r}\uparrow}^\dagger] = \sum_{\vec{r}'} [\epsilon_{\vec{r}\vec{r}'}^* - \mu \delta_{\vec{r}\vec{r}'}] \hat{c}_{\vec{r}'\uparrow}^\dagger - \Delta_{\vec{r}}^* \hat{c}_{\vec{r}\downarrow}. \quad \begin{array}{l} \text{Do commutation with } \vec{r}', \text{ then} \\ \text{change } \vec{r}' \text{ to } \vec{r}. \end{array} \quad (27.194a)$$

$$[\mathcal{H}_B, \hat{c}_{\vec{r}\downarrow}^\dagger] = \sum_{\vec{r}'} [\epsilon_{\vec{r}\vec{r}'}^* - \mu \delta_{\vec{r}\vec{r}'}] \hat{c}_{\vec{r}'\downarrow}^\dagger + \Delta_{\vec{r}}^* \hat{c}_{\vec{r}\uparrow} \quad (27.194b)$$

$$[\mathcal{H}_B, \hat{c}_{\vec{r}\uparrow}] = - \sum_{\vec{r}'} [\epsilon_{\vec{r}\vec{r}'} - \mu \delta_{\vec{r}\vec{r}'}] \hat{c}_{\vec{r}'\uparrow} + \Delta_{\vec{r}} \hat{c}_{\vec{r}\downarrow}^\dagger \quad \text{Take the complex conjugate of the first two relations.} \quad (27.194c)$$

$$[\mathcal{H}_B, \hat{c}_{\vec{r}\downarrow}] = - \sum_{\vec{r}'} [\epsilon_{\vec{r}\vec{r}'} - \mu \delta_{\vec{r}\vec{r}'}] \hat{c}_{\vec{r}'\downarrow} - \Delta_{\vec{r}} \hat{c}_{\vec{r}\uparrow}^\dagger. \quad (27.194d)$$

Placing the expression for $\hat{c}$ in terms of $\hat{\gamma}$ into any of these four expressions, one finds that the results are consistent only if the minus signs in relations between $\hat{c}$ and $\hat{\gamma}$ are chosen as in Eq. (27.192); matching terms in γ requires that

$$\begin{aligned} u_l(\vec{r}) \mathcal{E}_l &= \sum_{\vec{r}'} [\epsilon_{\vec{r}\vec{r}'} - \mu \delta_{\vec{r}\vec{r}'}] u_l(\vec{r}') + v_l(\vec{r}) \Delta_{\vec{r}} \\ v_l(\vec{r}) \mathcal{E}_l &= - \sum_{\vec{r}'} [\epsilon_{\vec{r}\vec{r}'}^* - \mu \delta_{\vec{r}\vec{r}'}^*] v_l(\vec{r}') + u_l(\vec{r}) \Delta_{\vec{r}}^*. \end{aligned} \quad (27.195)$$

These coupled eigenvalue equations are the *Bogoliubov equations*. The pair potential Δ is given by

$$\Delta_{\vec{r}} = \frac{U_0}{\mathcal{V}} N_k \langle \hat{c}_{\vec{r}\downarrow} \hat{c}_{\vec{r}\uparrow} \rangle = \sum_l \frac{U_0}{\mathcal{V}} u_l(\vec{r}) v_l^*(\vec{r}). \quad (27.196)$$

Interaction with Electromagnetic Fields. The reason to pose the Bogoliubov equations in real space is that external potentials can be incorporated into the matrix $\epsilon_{\vec{r}\vec{r}'}$, which also contains kinetic energy and contains effects of interaction with a lattice or impurities. For a homogeneous system, with no external potential, they reduce exactly to what appeared before in Eq. (27.168). The functions u_l and v_l are just plane waves:

$$u_{\vec{k}}^{(0)}(\vec{r}) = u_{\vec{k}} e^{-i\vec{k} \cdot \vec{r}}, \quad v_{\vec{k}}^{(0)}(\vec{r}) = v_{\vec{k}} e^{-i\vec{k} \cdot \vec{r}}. \quad \text{Identify the index } l \text{ with } \vec{k}. \quad (27.197)$$

Treating these solutions as the zero-order starting point, the Meissner effect emerges from finding effects of a weak external vector potential. The calculation is a rather elaborate exercise in perturbation theory.

As the vector potential turns on, the plane waves are altered only slightly, so the solutions of the perturbed problem can be indexed by $\vec{k}$, which takes the place of index l . The problem to be solved is

$$u_{\vec{k}}(\vec{r}) \mathcal{E}_{\vec{k}} = \left\{ \frac{1}{2m} \left(-i\hbar \vec{\nabla} + \frac{e\vec{A}}{c} \right)^2 - \mu \right\} u_{\vec{k}}(\vec{r}) + v_{\vec{k}}(\vec{r}) \Delta_{\vec{r}} \quad (27.198a)$$

$$v_{\vec{k}}(\vec{r}) \mathcal{E}_{\vec{k}} = - \left\{ \frac{1}{2m} \left(i\hbar \vec{\nabla} + \frac{e\vec{A}}{c} \right)^2 - \mu \right\} v_{\vec{k}}(\vec{r}) + u_{\vec{k}}(\vec{r}) \Delta_{\vec{r}}^*. \quad (27.198b)$$

Let

$$u_{\vec{k}}(\vec{r}) = u_{\vec{k}}^{(0)}(\vec{r}) + u_{\vec{k}}^{(1)}(\vec{r}) = u_{\vec{k}} e^{-i\vec{k} \cdot \vec{r}} + \sum_{\vec{k}'} e^{-i\vec{k}' \cdot \vec{r}} u_{\vec{k}}^{(1)}(\vec{k}') \quad (27.199a)$$

$$v_{\vec{k}}(\vec{r}) = v_{\vec{k}}^{(0)}(\vec{r}) + v_{\vec{k}}^{(1)}(\vec{r}) = v_{\vec{k}} e^{-i\vec{k} \cdot \vec{r}} + \sum_{\vec{k}'} e^{-i\vec{k}' \cdot \vec{r}} v_{\vec{k}}^{(1)}(\vec{k}'). \quad (27.199b)$$

Treating the vector potential as small, and collecting terms to first order, one has

$$\left(\mathcal{E}_{\vec{k}} - \frac{\hbar^2}{2m} \nabla^2 - \mu\right) v_{\vec{k}}^{(1)}(\vec{r}) - \Delta^* u_{\vec{k}}^{(1)}(\vec{r}) = -\frac{ie\hbar}{2mc} (\vec{A} \cdot \vec{\nabla} + \vec{\nabla} \cdot \vec{A}) v_{\vec{k}}^{(0)}(\vec{r}) \quad (27.200a)$$

$$\left(\mathcal{E}_{\vec{k}} + \frac{\hbar^2}{2m} \nabla^2 + \mu\right) u_{\vec{k}}^{(1)}(\vec{r}) - \Delta v_{\vec{k}}^{(1)}(\vec{r}) = -\frac{ie\hbar}{2mc} (\vec{A} \cdot \vec{\nabla} + \vec{\nabla} \cdot \vec{A}) u_{\vec{k}}^{(0)}(\vec{r}). \quad (27.200b)$$

In order to write down Eqs. (27.200), some technical problems have to be solved. The first is that the energy $\mathcal{E}_{\vec{k}} = \mathcal{E}_{\vec{k}}^{(0)} + \mathcal{E}_{\vec{k}}^{(1)}$ might change to first order in $\vec{A}$, but a change was not included. Problem 6 shows that $\mathcal{E}_{\vec{k}}^{(1)}$ vanishes so long as $\vec{\nabla} \cdot \vec{A} = 0$. In addition, $\Delta_{\vec{r}}$ has been replaced by the constant Δ , although $\Delta_{\vec{r}} = \Delta^{(0)} + \Delta_{\vec{r}}^{(1)}$ should in principle acquire a spatially varying contribution at first order. The demonstration is not straightforward, but $\Delta_{\vec{r}}^{(1)} = 0$ so long as the vector potential is restricted to the *London gauge*, $\vec{\nabla} \cdot \vec{A} = 0$. It is unfortunately not easy to carry out the full calculation in a fashion that is manifestly gauge-invariant.

Substituting Eqs. (27.199) into Eqs. (27.200) and integrating by $\int d\vec{r} \exp[i\vec{k}' \cdot \vec{r}] / \mathcal{V}$, one finds

$$(\mathcal{E}_{\vec{k}} + \zeta_{\vec{k}}) v_{\vec{k}}^{(1)}(\vec{k}') - \Delta^* u_{\vec{k}}^{(1)}(\vec{k}') = F_{\vec{k}'\vec{k}} v_{\vec{k}}^{(0)} \quad (27.201a)$$

$$(\mathcal{E}_{\vec{k}} - \zeta_{\vec{k}}) u_{\vec{k}}^{(1)}(\vec{k}') - \Delta v_{\vec{k}}^{(1)}(\vec{k}') = F_{\vec{k}'\vec{k}} u_{\vec{k}}^{(0)}, \quad (27.201b)$$

with the unperturbed single-particle energies

$$\zeta_{\vec{k}} = \frac{\hbar^2 k^2}{2m} - \mu, \text{ so that } \mathcal{E}_{\vec{k}} = \sqrt{\zeta_{\vec{k}}^2 + |\Delta|^2}, \quad (27.202)$$

and

$$F_{\vec{k}'\vec{k}} = -\frac{e\hbar}{2mc} \int \frac{d\vec{r}'}{\mathcal{V}} e^{i(\vec{k}' - \vec{k}) \cdot \vec{r}'} (\vec{k} + \vec{k}') \cdot \vec{A}(\vec{r}') = F_{\vec{k}\vec{k}'}^*. \quad (27.203)$$

One can immediately find the explicit expressions

$$v_{\vec{k}}^{(1)}(\vec{k}') = \frac{F_{\vec{k}'\vec{k}}}{\mathcal{E}_{\vec{k}}^2 - \mathcal{E}_{\vec{k}'}^2} [(\mathcal{E}_{\vec{k}} - \zeta_{\vec{k}}) v_{\vec{k}} + \Delta^* u_{\vec{k}}] \quad (27.204a)$$

$$u_{\vec{k}}^{(1)}(\vec{k}') = \frac{F_{\vec{k}'\vec{k}}}{\mathcal{E}_{\vec{k}}^2 - \mathcal{E}_{\vec{k}'}^2} [(\mathcal{E}_{\vec{k}} + \zeta_{\vec{k}}) u_{\vec{k}} + \Delta v_{\vec{k}}]. \quad (27.204b)$$

The perturbation theory is now finished, but there remains the task of finding the change in physical quantities of interest.

27.3.6 Derivation of Meissner Effect

To see whether the quantum wave function exhibits the Meissner effect, one must look at the response of the current to the vector potential. If the current has a linear response, then the material is superconducting. The current is

$$\vec{j} = -2e \frac{N_k}{\mathcal{V}} \text{Re} \left\langle \hat{c}_{\vec{r}\uparrow}^\dagger \left(\frac{\hat{p}}{m} + \frac{e\vec{A}}{mc} \right) \hat{c}_{\vec{r}\uparrow} \right\rangle \quad (27.205)$$

The factor of 2 is for spin, and the factor of $N_k/\mathcal{V}$ accounts for the fact that there are N_k values of $\vec{k}$, hence N_k values of $\vec{r}$ which occupy a total volume $\mathcal{V}$, so each $\vec{r}$ is associated with a volume $\mathcal{V}/N_k$.

$$= \frac{-e}{\mathcal{V}} \sum_{\vec{k}\vec{k}'} \left\langle \left(u_{\vec{k}'}^*(\vec{r}) \hat{\gamma}_{\vec{k}'\uparrow}^\dagger + v_{\vec{k}'}(\vec{r}) \hat{\gamma}_{\vec{k}\downarrow} \right) \left(\frac{\hat{p}}{m} + \frac{e\vec{A}}{mc} \right) \left(u_{\vec{k}}(\vec{r}) \hat{\gamma}_{\vec{k}\uparrow} + v_{\vec{k}}^*(\vec{r}) \hat{\gamma}_{\vec{k}\downarrow}^\dagger \right) \right\rangle$$

+ c.c. (27.206)

$$= \frac{-e}{\mathcal{V}} \sum_{\vec{k}} v_{\vec{k}}(\vec{r}) \left[\frac{\hbar \vec{\nabla}}{im} + \frac{e\vec{A}}{mc} \right] v_{\vec{k}}^*(\vec{r}) + \text{c.c.}$$

The only term that does not annihilate the ground state is $\hat{\gamma} \hat{\gamma}^\dagger$. (27.207)

Note that

$$\sum_{\vec{k}} v_{\vec{k}} v_{\vec{k}}^* = \frac{1}{2} \sum_{\vec{k}} \frac{\mathcal{E}_{\vec{k}} - \zeta_{\vec{k}}}{\mathcal{E}_{\vec{k}}} = N/2, \quad \text{Compare with Eqs. (27.145) and (27.173).} \quad (27.208)$$

so the current becomes

$$\vec{j} = \vec{j}^1 - \frac{ne^2 \vec{A}}{mc}, \quad n \text{ is the total density of electrons.} \quad (27.209)$$

where

$$\vec{j}^1 = -e \frac{1}{\mathcal{V}} \sum_{\vec{k}} v_{\vec{k}} \frac{\hbar \vec{\nabla}}{im} v_{\vec{k}}^*(\vec{r}) + \text{c.c.} \quad (27.210)$$

$\vec{j}^1$ requires further evaluation. To first order in $\vec{A}$,

$$\vec{j}^1 = -\frac{e\hbar}{\mathcal{V}m} \sum_{\vec{k}\vec{k}'} v_{\vec{k}} v_{\vec{k}}^{(1)*}(\vec{k}') (\vec{k} + \vec{k}') e^{i(\vec{k}' - \vec{k}) \cdot \vec{r}} + v_{\vec{k}}^* v_{\vec{k}}^{(1)}(\vec{k}') (\vec{k} + \vec{k}') e^{-i(\vec{k}' - \vec{k}) \cdot \vec{r}}. \quad (27.211)$$

Substituting the expression for $v_{\vec{k}}^{(1)}(\vec{k}')$ obtained in Eq. (27.204) and using Eq. (27.173) gives that

$$\vec{j}^1 = \frac{-e\hbar}{m\mathcal{V}} \sum_{\vec{k}\vec{k}'} \left(\vec{k} + \vec{k}' \right) e^{i(\vec{k}' - \vec{k}) \cdot \vec{r}} \frac{F_{\vec{k}'\vec{k}}}{\mathcal{E}_{\vec{k}}^2 - \mathcal{E}_{\vec{k}'}^2} \left[(\mathcal{E}_{\vec{k}} - \zeta_{\vec{k}'}') \left(\frac{\mathcal{E}_{\vec{k}} - \zeta_{\vec{k}}}{2\mathcal{E}_{\vec{k}}} \right) + \frac{\Delta^* \Delta}{2\mathcal{E}_{\vec{k}}} \right]$$

$$+ \left(\vec{k} + \vec{k}' \right) e^{-i(\vec{k}' - \vec{k}) \cdot \vec{r}} \frac{F_{\vec{k}'\vec{k}}^*}{\mathcal{E}_{\vec{k}}^2 - \mathcal{E}_{\vec{k}'}^2} \left[(\mathcal{E}_{\vec{k}} - \zeta_{\vec{k}}) \left(\frac{\mathcal{E}_{\vec{k}} - \zeta_{\vec{k}}}{2\mathcal{E}_{\vec{k}}} \right) + \frac{\Delta \Delta^*}{2\mathcal{E}_{\vec{k}}} \right] \quad (27.212)$$

$$= \frac{-e\hbar}{m\mathcal{V}} \sum_{\vec{k}\vec{k}'} \left(\vec{k} + \vec{k}' \right) e^{i(\vec{k}' - \vec{k}) \cdot \vec{r}} F_{\vec{k}'\vec{k}} L(\zeta_{\vec{k}}, \zeta_{\vec{k}'}'), \quad \text{Interchanging } \vec{k} \text{ and } \vec{k}' \text{ in the second half of the right-hand side and doing a moderate amount of algebra.} \quad (27.213)$$

where

$$L(\zeta_{\vec{k}}, \zeta_{\vec{k}'}') = \frac{\mathcal{E}_{\vec{k}} \mathcal{E}_{\vec{k}'}' - \zeta_{\vec{k}} \zeta_{\vec{k}'}' - \Delta^* \Delta}{2(\mathcal{E}_{\vec{k}} + \mathcal{E}_{\vec{k}'}') \mathcal{E}_{\vec{k}} \mathcal{E}_{\vec{k}'}'}. \quad (27.214)$$

Because L depends upon the absolute value of $\vec{k}$ and $\vec{k}'$ only through the energies $\zeta_{\vec{k}}$, one can do the angular integrals over $\vec{k}$ and $\vec{k}'$. Define

$$\sigma_{\alpha\beta}(\zeta, \zeta', \vec{R}) = \frac{2\pi\hbar}{2\mathcal{V}^2} \left(\frac{e\hbar}{m} \right)^2 \sum_{\vec{k}\vec{k}'} \delta(\zeta_{\vec{k}} - \zeta) \delta(\zeta_{\vec{k}'} - \zeta') (k_{\alpha} + k'_{\alpha}) (k_{\beta} + k'_{\beta}) e^{i(\vec{k} - \vec{k}') \cdot \vec{R}}. \quad (27.215)$$

Then expanding out $F_{\vec{k}'\vec{k}}$ using Eq. (27.203) gives

$$\vec{j}_\alpha^1(\vec{r}) = \frac{1}{2\pi\hbar c} \sum_\beta \int d\vec{r}' d\zeta d\zeta' L(\zeta, \zeta') \sigma_{\alpha\beta}(\zeta, \zeta', \vec{r} - \vec{r}') A_\beta(\vec{r}'). \quad (27.216)$$

The conductivity σ can be evaluated by defining

$$Q(\zeta, \vec{r}) = \frac{1}{V} \sum_{\vec{k}} \delta(\zeta_{\vec{k}} - \zeta) e^{-i\vec{k}\cdot\vec{r}} \approx \frac{D(\mathcal{E}_F) \sin \sqrt{2m\zeta/\hbar^2} r}{2k_F r}, \quad (27.217)$$

Evaluate energy and wave vector at the Fermi surface, because this is the only place where the coherence factor $L(\zeta, \zeta')$ is much different from zero.

so that

$$\sigma_{\alpha\beta}(\zeta, \zeta', \vec{R}) = -\frac{2\pi\hbar}{2} \left(\frac{e\hbar}{m} \right)^2 \frac{\partial}{\partial a_\alpha} \frac{\partial}{\partial a_\beta} Q(\zeta, \vec{R} - \vec{a}) Q(\zeta', -(\vec{R} + \vec{a})) \Big|_{\vec{a}=0} \quad (27.218)$$

$$\approx e^2 \frac{v_F}{2\pi} D(\mathcal{E}_F) \frac{R_\alpha R_\beta}{R^4} \cos \left[\frac{(\zeta - \zeta')R}{\hbar v_F} \right]. \quad (27.219)$$

In taking the derivatives with respect to a_α or a_β , keep only the largest powers of $k_F R$, which is much larger than 1 for most important values of R .

Therefore, the first contribution to the current can now be written

$$j_\alpha^1(\vec{r}) = \sum_\beta \int d\vec{r}' S_{\alpha\beta}^1(\vec{r} - \vec{r}') A_\beta(\vec{r}'), \quad (27.220)$$

with

$$S_{\alpha\beta}^1(\vec{R}) = \frac{3ne^2}{4\pi^2 m \hbar v_F} \frac{R_\alpha R_\beta}{R^4} \int d\zeta d\zeta' L(\zeta, \zeta') \cos \left[\frac{(\zeta - \zeta')R}{\hbar v_F} \right]. \quad (27.221)$$

The coherence length ξ is conventionally defined for the microscopic model as

$$\xi = \frac{\hbar v_F}{\pi \Delta}. \quad (27.222)$$

Upon evaluating the final integral in Eq. (27.221) (Problem 7) and summing both terms in Eq. (27.209) the current is

$$j_\alpha(\vec{r}) = \sum_\beta \int d\vec{r}' S_{\alpha\beta}(\vec{r} - \vec{r}') A_\beta(\vec{r}') \quad (27.223a)$$

with

$$S_{\alpha\beta}(\vec{R}) = \frac{-3ne^2}{4\pi m c \xi} \frac{R_\alpha R_\beta}{R^4} I(R) \quad (27.223b)$$

and

$$I(R) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{dy'}{\cosh y'} \exp \left[-\frac{2R}{\pi \xi} \cosh y' \right]. \quad (27.223c)$$

When $\vec{A}$ varies slowly on the scale of the coherence length ξ , it can be taken out of the spatial integral, and one finds that

$$\vec{j}(\vec{r}) = \frac{-ne^2}{mc} \vec{A}(\vec{r}). \quad (27.224)$$

If $\vec{A}$ does not vary slowly, Eq. (27.223a) describes averages of $\vec{A}$ over a ball the size of the coherence length ξ .

In either case, the microscopic theory predicts perfect diamagnetism. For example, operating with $4\pi/c \vec{\nabla} \times$ on both sides of Eq. (27.224) reproduces the London equation Eq. (27.5). More generally, the relation between current and vector potential in Eq. (27.223) meets the conditions laid down in Section 27.2.1 for production of the Meissner effect. The microscopic theory has predicted that superconductors will be perfect diamagnets, and it also shows how the coherence length ξ is related to the gap parameter Δ .

27.3.7 Comparison with Experiment

According to Bardeen, "In working out the properties of our simplified model and comparing with experimental results on real metals, the excellent agreement obtained was a continual source of amazement Everything fitted together neatly like the pieces of a jigsaw puzzle. Accordingly, the scepticism with which the theory was greeted in some quarters was surprising." [Bardeen (1963), p. 7]. The resistance was rapidly conquered by several comparisons with experiment, including specific heat and nuclear spin relation. The predictions of how these two effects should vary with temperature was computed by methods similar to those used for the Meissner effect. Specific heat is obtained by differentiating the expectation value of the Hamiltonian with respect to temperature, while spin relaxation involves the expectation value $\langle \hat{c}_\uparrow^\dagger \hat{c}_\downarrow \rangle$. The forms of temperature dependence are different below T_c . Specific heat drops off rapidly as T drops below T_c , while spin relaxation at first rises when T goes down, eventually dropping to zero. These two predictions were confirmed experimentally, as shown in Figure 27.11, and played a decisive role in confirming the correctness of the basic model.

Improvements in photoemission have made it possible to observe directly changes in the electron density of states due to the superconducting transition. The density of electronic states in BCS theory should be

$$D(E_k) = \left(\frac{d\mathcal{E}_k}{dk} \right)^{-1} = \left(\frac{d\mathcal{E}_k}{d\epsilon_k} \right)^{-1} \left(\frac{d\epsilon_k}{dk} \right)^{-1} = D(\mathcal{E}_F) \operatorname{Re} \frac{|\mathcal{E}_k|}{\sqrt{\mathcal{E}_k^2 - \Delta^2}}. \quad \text{Using Eq. (27.142)} \quad (27.225)$$

Eq. (27.225) predicts that the density of states vanishes completely within a range $\pm\Delta$ around $\mathcal{E}_F$. This is not observed in photoemission experiments. Instead the density of states is smeared out by the finite lifetime of excitations. Dynes et al. (1978) proposed a simple modification of Eq. (27.225), adding an imaginary part

to the energy, so that the probability of interacting with a state of energy $\mathcal{E}$ becomes

$$f(\mathcal{E})D(\mathcal{E}) = f(\mathcal{E})\text{Re} \frac{\mathcal{E} - i\Gamma}{\sqrt{\mathcal{E} - i\Gamma + \Delta}\sqrt{\mathcal{E} - i\Gamma - \Delta}}. \quad (27.226)$$

$f(\mathcal{E})$ is the Fermi function. The expression is set up to choose the correct branch of the square roots. Γ functions as an extra fitting parameter.

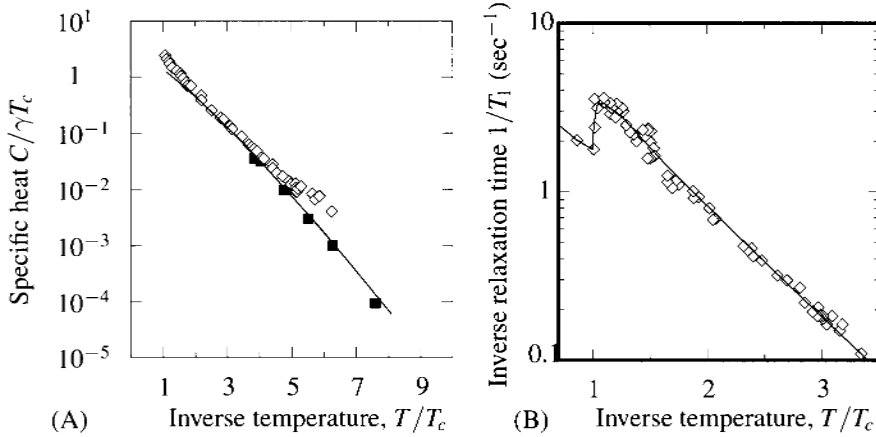


Figure 27.11. (A) Specific heat of aluminum and vanadium, relative to γT_c , where γ is the Sommerfeld parameter defined in Eq. (6.78), compared to the prediction of Bardeen, Cooper, and Schrieffer. [Source: Boorse (1959), p. 391.] (B) Inverse nuclear spin relaxation in aluminum compared with prediction of Bardeen, Cooper, and Schrieffer. Note the initial rise below T_c followed by an exponential drop. [Source: Masuda and Redfield (1962), p. 161.]

Some questions could not, however, be answered from the model Hamiltonian alone. So long as the effective electron–electron interaction $U_{\vec{k}}$ is negative, the model always predicts superconductivity, leaving much room for a more detailed explanation of which materials are superconducting, which are not, and what sets the critical temperature. To address such questions it is necessary to return to Eq. (27.104). A thorough discussion has been provided by McMillan (1968), building upon work of Eliashberg (1960). The calculations are very elaborate, but the flavor of what is accomplished can be captured by letting angular brackets denote averages with respect to $\vec{k}$ and $\vec{k}'$ over the Fermi surface, take $\vec{q} = \vec{k} - \vec{k}'$ and defining

$$\lambda_{\text{ep}} = -D(\mathcal{E}_F) \left\langle \frac{4\pi e^2}{q^2 + \kappa_c^2} \frac{\omega_{\vec{q}}^2 - \bar{\omega}_{\vec{q}}^2}{(\epsilon_{\vec{k}} - \epsilon_{\vec{k}'})^2 / \hbar^2 - \omega_{\vec{q}}^2} \right\rangle, \quad \mu^* = D(\mathcal{E}_F) \left\langle \frac{4\pi e^2}{q^2 + \kappa_c^2} \right\rangle. \quad (27.227)$$

Thus μ^* is related to a product of the strength of the Coulomb interaction and the density of states at the Fermi surface, while λ_{ep} is roughly the electron–phonon interaction times the density of states at the Fermi surface. Physical quantities needed to evaluate these averages include the following:

- Electron energy bands near the Fermi surface.

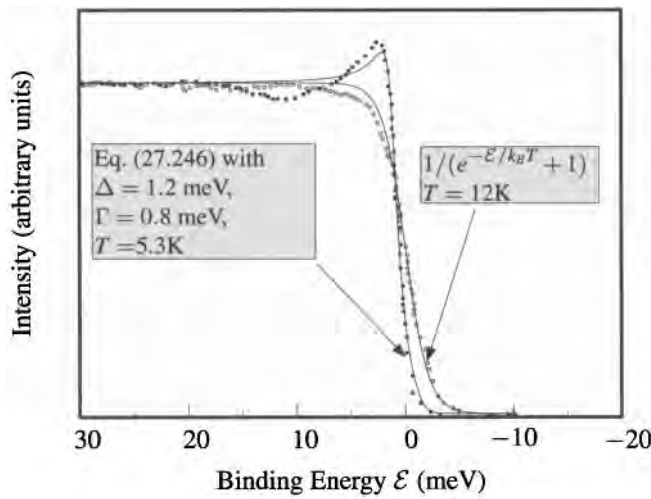


Figure 27.12. High-resolution photoemission from lead, above and below the superconducting transition. For temperatures $T > T_c$, the photoemission intensity is a Fermi function. Below T_c , the data are described by Eq. (27.226). Lines are theoretical expressions and circles are data. [Source: Yokoya et al. (2002), p. 102.]

- Phonon densities of states.
- Electron–phonon interaction matrix elements.

Following a sophisticated numerical analysis, McMillan finds that the superconducting transition temperature T_c is given approximately by

$$T_c = \frac{\Theta_D}{1.45} \exp \left\{ - \left[\frac{(1 + \lambda_{ep})}{\lambda_{ep} - \mu^* (1 + 0.62 \lambda_{ep})} \right] \right\}, \quad (27.228)$$

with Θ_D the Debye temperature. According to Table 13.1, Debye temperatures are on the order of a few hundred kelvin. Following Eq. (22.42), one can estimate the order of magnitude of the electron–phonon coupling as $C \approx \sqrt{2\pi e^2 \hbar \omega / k_F^2}$. Using also Eq. (6.33) to estimate the density of states at the Fermi surface, and taking $\vec{k} = \vec{k}'$, gives finally the very rough estimate $\lambda_{ep} \approx 0.8 \cdot [10^{23} \text{ cm}^{-3} / n]^{1/3}$. A value of 0.8 for λ_{ep} should be an upper limit, because the average in Eq. (27.227) involves regions that are both positive and negative, and cancel against each other. Neglecting $\mu^* \approx 0.1$ and taking $\Theta_D = 400$ K gives a superconducting transition temperature of around 30 K. For many years, Nb_3Ge held the record as material with the highest superconducting transition temperature, with $T_c = 23$ K, and 30 K was widely suspected to be a theoretical upper bound.

27.3.8 High-Temperature Superconductors

Bednorz and Müller (1986) found that $\text{La}_{2-x}\text{Ba}_x\text{CuO}_4$ (LBCO) becomes superconducting at a temperature of 35 K, triggering a frenzied hunt for new superconduct-

ing materials. Soon afterwards Wu et al. (1987) found $\text{YBa}_2\text{Cu}_3\text{O}_{6+x}$ (1–2–3 compound, or YBCO) with a transition temperature of 92K, meaning that it could be driven superconducting by immersion in liquid nitrogen at 77K. Additional compounds were eventually found with superconducting transitions at temperatures over 100 K. The *high-temperature superconductors* are different from the compounds that set the previous records. They are not conventional metals. Instead, they are antiferromagnetic insulators, carefully doped so as to produce metallic and superconducting phases. Many of them are based upon layers of CuO_2 ; just as with CuO (Section 23.6.3) the insulating behavior is induced by electron correlations and cannot be explained in the single-electron picture. While there is no full consensus on the theoretical description of high-temperature superconductors, many experimental claims are now fairly well established.

Structure. The high-temperature superconductors are brittle ceramics. Figure 27.13 shows the structure of a well-studied member of the family, $\text{La}_{2-x}\text{Sr}_x\text{CuO}_4$, also called *LSCO* or *La214*. The superconductivity is due to motion of electrons on the copper-oxygen planes. The unit in the center of the crystal is perovskite (Section 2.3.6), explaining why the copper-oxide superconductors are also called perovskites. The crystal has both tetragonal and orthorhombic phases, which result from slight symmetry-breaking distortions of the structure shown in the figure.

The parent compound, La_2CuO_4 is an antiferromagnetic insulator. Superconductors are produced by substituting varying degrees of strontium for the lanthanum. Lanthanum has a single *d* electron in an outer shell, while strontium has a filled *s* shell, so the effect of substituting strontium for lanthanum is to dope the structure with holes. Some fraction of the positive charge from the holes makes its way to the copper-oxygen planes, and in range of concentrations produces superconductivity. The maximum T_c of 38K is reached for $x \approx 0.15$.

Phase Diagram.

The full array of experimental probes in condensed matter physics has been brought to bear on the high-temperature superconductors. One of the central conclusions is that the phase diagrams of a large number of the materials are essentially identical, once they have been scaled by the maximum transition temperature, and by the optimal dopant concentration. The result appears in Figure 27.14.

Antiferromagnetic Region When dopant concentration x is zero, all of the copper-oxide ceramic superconductors are anti-ferromagnetic insulators. This is remarkable. Superconductivity consists in the expulsion of magnetic flux from a solid, so magnetic ordering and superconductivity are naturally considered to be competing and incompatible forms of order. The Curie temperatures where antiferromagnetic order disappears are on the order of several hundred degrees Kelvin.

Superconducting Region Figure 27.14 shows that just when the doping x reaches the point where the antiferromagnetic transition temperature T_c drops to zero, superconductivity appears. As the dopant concentration x increases, and the hole

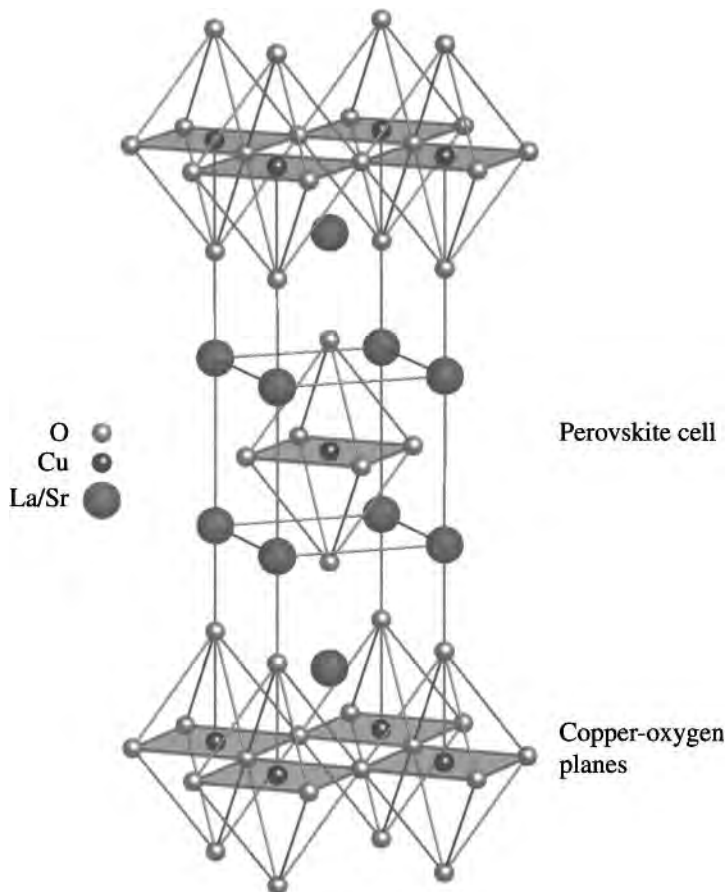


Figure 27.13. Structure of La_2CuO_4 , as reported by Yamada et al. (1988). There are both tetragonal and orthorhombic phases, which however cannot be distinguished on the scale of this figure. Superconductivity results from substitution of Sr for La to form $\text{La}_{2-x}\text{Sr}_x\text{CuO}_4$

density available to the copper-oxygen planes increases, and the superconducting transition temperature increases, reaching a maximum around $x = .15$ for LSCO and $x = .18$ for $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+x}$ (Bi2212). Then the superconducting transition temperature decreases again, reaching 0 at around $x = 0.3$. For higher dopant concentrations, the ceramics are normal metals.

The coherence length ξ of the high-temperature superconductors is anisotropic, but small, on the order of 1 or 2 Å. The parameter κ defined in Eq. (27.44) is much larger than 1, so these superconductors are of type II. The usefulness of the materials would greatly be increased if they continued to act as superconductors for magnetic fields greater than H_{c2} where vortices first form. However, the small coherence length leads naturally to large numbers of vortices with narrow cores that are very hard to pin, and drift of the vortices creates dissipation.

Pseudogap Region Figure 27.14 shows another dividing line, detected in the high-temperature superconductors by a wide variety of probes. The dividing line is a crossover temperature T^* , as no thermodynamic quantity has yet shown an abrupt enough change to qualify as a true phase transition. However, below this line, there are many unusual properties. The region is often called the *pseudogap* phase

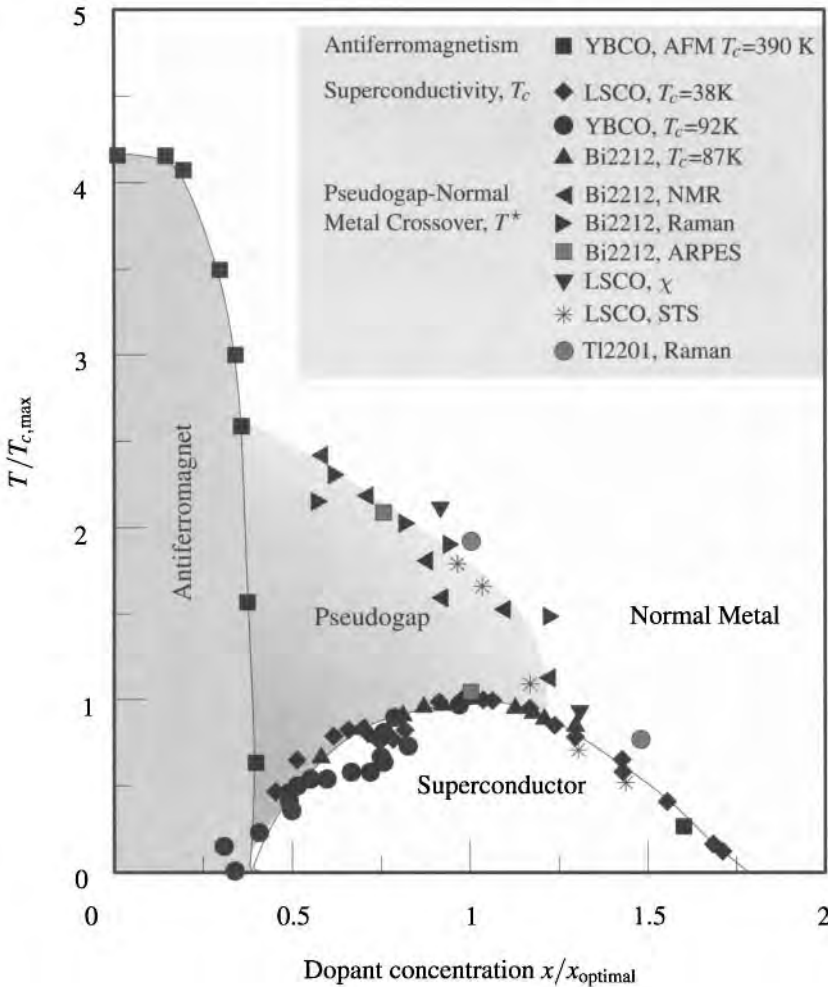


Figure 27.14. Phase diagram for a number of copper-oxide superconductors. The horizontal axis is scaled by the optimal hole concentration, which is on the order of 0.2 per unit cell. The vertical temperature axis is scaled by the maximum critical superconducting temperature, typically around 100 K. Scaled in this way, the phase diagrams of many different compounds collapse onto each other. Bi2212 is $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+x}$ and Tl2201 is $\text{Tl}_2\text{Ba}_2\text{CuO}_{6+x}$. The crossover temperature T^* is detected by many different probes including nuclear magnetic resonance (NMR), Raman scattering, angle-resolved photoemission spectroscopy (ARPES), magnetic susceptibility (χ), and scanning tunneling spectroscopy (STS). [Source: Rossat-Mignod et al. (1990), Greene and Bagley (1990), p. 529 and Nakano et al. (1998) p. 2623.]

because as discussed by Damascelli et al. (2003), angle-resolved photoemission detects a superconducting gap, by techniques like that shown in Figure 27.12, along some but not all directions perpendicular to the Fermi surface. Investigations with scanning tunneling probes, reviewed by Fischer et al. (2007), tell a similar story. Almost any probe one chooses shows some sort of cross-over between two limiting behaviors at round T^* , and thus one can map out the line with Raman spectroscopy, nuclear magnetic resonance, or magnetic susceptibility as well.

Even the normal metal phase may be unusual. Electrical resistivity is surprisingly linear in temperature; in the case of YBCO, the linear behavior persists up to around 1000 K. The various competing contributions to resistivity, as for example in Figure 18.1, are mysteriously absent. A phenomenological account of this behavior is provided by Varma et al. (1989).

Theories While experimentalists have been busy mapping out the phase diagram, theorists have been busy trying to calculate it. No definitive answer has yet been reached. A very appealing theory is that the essence of the high-temperature superconductors is to be found in the Hubbard model of Section 26.7. The high-temperature superconductors are no more and no less than doped Mott-Hubbard insulators. The challenge faced by this point of view is that it has so far not proved possible to solve the Hubbard model exactly, nor has any method of approximation been found completely convincing. Thus, if some approximate solution is found that reproduces some feature of the experiments, it is hard to tell whether this feature really resulted from the Hubbard model itself, or from the choice of approximations. Lee et al. (2006) provide a confident case that Mott-Hubbard physics does indeed explain the main features of the copper-oxide ceramic superconductors. Chen et al. (2005) provide a careful presentation of some alternatives. If one theory does eventually emerge dominant, it will happen because the great majority of experimentalists finds it useful in interpreting their experiments. This has not yet happened with a single microscopic theory.

Symmetry of the Order Parameter.

The majority of experimental evidence supports the claim that the superconducting order parameter of the high T_c materials has d -wave character, while for most previous superconductors it had s -wave character. Before displaying the experimental evidence, it is necessary to establish what this claim means, which requires in turn asking about the relationship between the Landau–Ginzburg equation and the underlying microscopic theory of Bardeen, Cooper, and Schrieffer.

The theory of Landau and Ginzburg is built upon the order parameter Ψ . The most important properties of this order parameter are, first, that it becomes nonzero only in the presence of superconductivity and, second, that it alters under gauge transformations

$$\vec{A} \rightarrow \vec{A} + \vec{\nabla} \chi \text{ as } \Psi \rightarrow \Psi e^{-2ie\chi/\hbar c}. \quad (27.229)$$

The phase of Ψ then naturally satisfied the conditions outlined in Section 27.2.9 that lead to the Josephson effect.

The Bogoliubov equations feature a quantity that shares these two properties—the pair potential $\Delta_{\vec{r}}$. To see why, turn to Eqs. (27.198), and ask how u , v , and Δ transform after gauge transformations. To keep the form of the equations intact, they transform as

$$u_{\vec{k}}(\vec{r}) \rightarrow u_{\vec{k}}(\vec{r})e^{-ie\chi/\hbar c}, \quad v_{\vec{k}}(\vec{r}) \rightarrow v_{\vec{k}}(\vec{r})e^{ie\chi/\hbar c}, \quad \text{and} \quad \Delta_{\vec{r}} \rightarrow \Delta_{\vec{r}}e^{-2ie\chi/\hbar c}. \quad (27.230)$$

Very near the transition temperature T_c , Gor'kov (1959) showed that $\Delta_{\vec{r}}$ is in fact proportional to Ψ . At lower temperatures the correspondence is less certain, but the precise form of the Landau–Ginzburg equations is not necessarily correct in any event.

In searching for the form that a theory of high-temperature superconductivity might take, it is natural to ask how $\Delta_{\vec{r}}$ might generalize when the simplifying assumptions leading to Eq. (27.196) no longer apply. If the effective potential U is not isotropic, then

$$\Delta_{\vec{k}\vec{q}} = \sum_{\vec{k}'} \frac{U_{\vec{k}\vec{k}'}}{\mathcal{V}} \langle \hat{c}_{\vec{q}-\vec{k}'\downarrow} \hat{c}_{\vec{k}'\uparrow} \rangle. \quad (27.231)$$

Fourier transforming according to prescriptions as in Eq. (27.189) gives

$$\Delta_{\vec{r}\vec{r}'} = \sum_{\vec{r}''} \frac{U_{\vec{r}\vec{r}''}}{\mathcal{V}} \langle \hat{c}_{-\vec{r}''\downarrow} \hat{c}_{\vec{r}''-\vec{r}'\uparrow} \rangle. \quad (27.232)$$

The implication of this calculation is that in a general theory of superconductivity, the analog of Ψ should depend upon two arguments, becoming $\Psi(\vec{r}, \vec{r}')$. The calculations worked out so far assumed that the effective potential U was isotropic, and consequently that Ψ depended only upon the center-of-mass coordinate $(\vec{r} + \vec{r}')/2$. In a more general theory, $\Psi(\vec{r}, \vec{r}')$ might be expected to do the following:

$$\text{Vary slowly as a function of } (\vec{r} + \vec{r}')/2. \quad (27.233a)$$

$$\text{Decay rapidly as a function } R = |\vec{r} - \vec{r}'|. \quad (27.233b)$$

$$\text{Depend upon the direction of } \vec{R} = \vec{r} - \vec{r}'. \quad (27.233c)$$

A superconductor where Ψ is independent of the direction of $\vec{R}$ is called *s-wave*. One where Ψ has the symmetry of $\vec{x} \cdot \vec{R} \propto \cos \theta$ is *p-wave*. Superfluid ^3He has an order parameter of this type. One with the symmetry of $(\vec{x} \cdot \vec{R})^2 - (\vec{y} \cdot \vec{R})^2 \propto \cos 2\theta$ is called *d-wave*. Problem 8 shows that under some simplifying assumptions, one should expect to measure

$$|\Delta_{\vec{k}}| \propto |\cos 2\phi| \quad \text{Where } \phi \text{ is the angle of the vector } (k_x, k_y). \quad (27.234)$$

Experimental investigations of the symmetry of the order parameter can be carried out without commitment to any particular microscopic theory of the interaction U , as discussed by Annett et al. (1996). Two particularly direct experiments provide evidence for *d-wave* pairing. The first is Josephson tunneling into superconducting samples from different angles, as performed by Wollman et al. (1995).

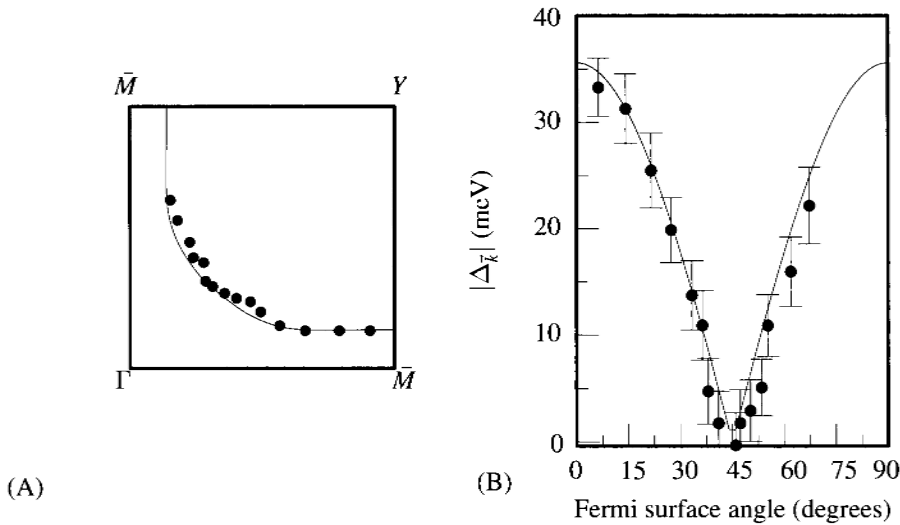


Figure 27.15. Experimental evidence for d -wave pairing in the high-temperature superconducting ceramic $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$ (Bi2212). (A) Experimental trace of the Fermi surface at 13K obtained from angle-resolved photoemission. (B) Measurement of the superconducting gap $\Delta_{\vec{k}}$ as a function of angle. The solid line is a plot of $|\cos(ak_x) - \cos(ak_y)|$. [Source: Ding et al. (1996), p. 799.]

Even more direct and convincing are results from angle-resolved photoemission spectroscopy, discussed by Damascelli et al. (2003). The methods employed are those shown in Figure 27.12. Opening a superconducting gap creates a small bump in the density of states below the Fermi surface, and the size and shape of the bump can be used to fit to the magnitude of the superconducting gap $|\Delta_{\vec{k}}|$. Figure 27.15 shows direct measurements of variations in the size of the superconducting gap $|\Delta_{\vec{k}}|$ consistent with Eq. (27.234).

Thus, a microscopic theory of high-temperature superconductivity needs to explain the coexistence of superconductivity and antiferromagnetism in the same material, reproduce the main features of the phase diagram in fig:ybco.thermo, and incorporate d -wave symmetry. The problem is not that there is no theory of the high-temperature superconducting materials. The problem is that there are many, and none has yet carried the day. The last word belongs to Anderson (1997): “the consensus is that there is absolutely no consensus on the theory of high T_c superconductivity.”

To those men of early times and, as it were, first parents of philosophy, to Aristotle, Theophrastus, Ptolemaeus, Hippocrates, Galen, be due honor rendered ever, for from them has knowledge descended to those that have come after them: but our age has discovered and brought to light very many things which they too, were they among the living, would cheerfully adopt. Wherefore we have had no hesitation in setting forth in hypothe-

ses that are provable, the things that we have through long experience discovered. Farewell.

— Gilbert (1600), p. li

Problems

1. **Superconducting sphere:** Consider a sphere of superconducting material of type I, whose critical field is H_c , in a uniform external magnetic field $\vec{H}$.

- (a) Using the fact that normal magnetic fields must vanish at the surface of a superconductor, show that when the external field H reaches a value of $\frac{2}{3}H_c$, some portion of the sphere must make the transition to normal metal. Do not try to find the shape of the region that does so.
- (b) At what external field will the entire sphere become normal?

2. **Energy of normal-superconducting interfaces:** Consider a half-space, $x < 0$, of normal metal in equilibrium with a superconductor filling $x > 0$. To allow the normal and superconducting metals to remain in equilibrium, the normal metal is permeated almost everywhere by a magnetic induction B_c .

- (a) First consider the limit $\kappa \rightarrow 0$, where the penetration depth λ_L is small. Because superconductivity vanishes in the normal metal, it is reasonable to guess that one needs to look for a solution in the superconductor such that Ψ vanishes at $x = 0$, such as Eq. (27.41). Find the difference in free energy $\tilde{\mathcal{G}}$ between a superconductor where Ψ vanishes at $x = 0$, and one where the wave function Ψ fills the space $x > 0$ uniformly. Show that the interface energy is positive, and given by Eq. (27.52).
- (b) Next consider the opposite limit, in which the coherence length ξ is small compared to the penetration depth λ_L . Now the gradients of Ψ can be neglected, and instead one has to consider the contributions of the magnetic field. First compute the free energy per area

$$\mathcal{F} = \int_0^\infty dx \Psi_0^2 \frac{4e^2}{2mc^2} A^2 + \frac{B^2}{8\pi}. \quad (27.235)$$

Simplify $\mathcal{F}$ with the use of Eq. (27.9). Then compute $\tilde{\mathcal{G}}$ as defined in Eq. (24.27), and verify Eq. (27.53).

3. **Diffraction effects in Josephson junctions:** Referring to Figure 27.5, and employing Eq. (27.68a), show that the maximum zero-voltage current J_c able to flow through a rectangular Josephson junction in the presence of a magnetic field is given by

$$J_c \propto \left| \frac{\sin(2\pi\Phi/\Phi_0)}{(2\pi\Phi/\Phi_0)} \right|. \quad (27.236)$$

- (a) Write down a vector potential $\vec{A}$ that produces the desired field $\vec{B}$

(b) Assuming some phases ϕ_1 and ϕ_2 on the two sides of the junction, find the total current J by integrating (27.68a) across the area of the junction.

(c) Choose the phases ϕ_1 and ϕ_2 to maximize the zero-voltage current flow.

4. **Josephson Lagrangian:** Verify Eqs. (27.84).

5. **Properties of BCS coherent state:** Verify Eqs. (27.130) through (27.133).

6. **London gauge:**

(a) Show that if $\vec{\nabla} \cdot \vec{A} = 0$, then first-order changes in the energies $\mathcal{E}_{\vec{k}}$ do not appear in Eqs. (27.200).

(b) Verify that $\Delta_{\vec{r}}$ responds to a change in gauge χ as reported in Eq. (27.230).

(c) Physical properties of superconductors are functionals of the vector potential $\vec{A}$. For small vector potentials, the functional is linear, and in a homogeneous superconductor one must have

$$\Delta_{\vec{r}} = \Delta^{(0)} + \Delta_{\vec{r}}^{(1)} = \Delta^{(0)} + \int d\vec{r}' \vec{\mathcal{P}}(\vec{r} - \vec{r}') \cdot \vec{A}(\vec{r}'), \quad (27.237)$$

for some function $\vec{\mathcal{P}}$. Consider the particular case where $\vec{A} = \vec{\nabla}\chi$, and χ is very small. Linearizing Eq. (27.230) and comparing it with Eq. (27.237), show that

$$\vec{\mathcal{P}}(\vec{r}) = \frac{-ie\Delta^{(0)}}{2\pi\hbar c} \vec{\nabla} \left(\frac{1}{r} \right). \quad (27.238)$$

(d) Show as a consequence that in the London gauge, where $\vec{\nabla} \cdot \vec{A} = 0$, and if $\vec{A} \cdot \hat{n}$ vanishes at the outer reaches of the system, with $\hat{n}$ is a unit normal, then $\Delta_{\vec{r}}^{(1)} = 0$, and $\Delta_{\vec{r}} = \Delta$. In other words, gauge invariance is used to choose a convenient gauge in which the problem simplifies.

7. **Meissner effect integral:** Evaluate

$$W(R) \equiv \int d\zeta d\zeta' L(\zeta, \zeta') \cos [(\zeta - \zeta')R/\hbar v_F]. \quad (27.239)$$

(a) Use Eq. (27.214) for $L(\zeta, \zeta')$ and Eq. (27.202) for $\mathcal{E}_{\vec{k}}$. Make the substitutions $\zeta = \Delta \sinh x$ and $\zeta' = \Delta \sinh x'$.

(b) Define $y = (x - x')/2$ and $y' = (x + x')/2$. Express the integral in terms of y and y' .

(c) Retain the variable y' , but make the substitution $p = \sinh y$. The integral over p can be performed to give

$$W(R) = \pi \hbar v_F \delta(R) - \pi \Delta \int dy' \frac{\exp[-2\Delta \cosh y'R/\hbar v_F]}{\cosh y'}. \quad (27.240)$$

8. **d -wave Superconductivity:** Suppose that the gap function $\Delta_{\vec{r}\vec{r}'}$ of Eq. (27.232) obeys the conditions of Eqs. (27.233); in fact, there is no dependence upon $(\vec{r} + \vec{r}')$. Suppose that the dependence on the direction of $\vec{R}$ is of the d -wave form

$$(\vec{x} \cdot \vec{R})^2 - (\vec{y} \cdot \vec{R})^2 \propto \cos 2\theta. \quad (27.241)$$

Show that $\Delta_{\vec{k}\vec{k}'}$ takes the form for some function $f(k)$

$$\Delta_{\vec{k}\vec{k}'} = \delta_{\vec{k}\vec{k}'} \left(\frac{\partial^2}{\partial k_x^2} - \frac{\partial^2}{\partial k_y^2} \right) f(k), \quad (27.242)$$

resulting in a symmetry for the gap function given by Eq. (27.234).

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